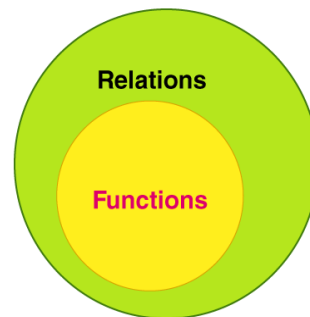
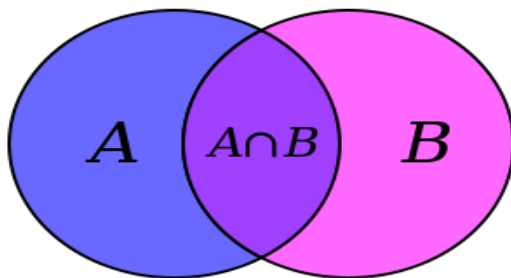


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DISCRETE MATHEMATICS



p	q	$\neg p$	$\neg q$	$p \wedge q$	$p \vee q$	$p \Rightarrow q$	$p \Leftrightarrow q$
T	T	F	F	T	T	T	T
T	F	F	T	F	T	F	F
F	T	T	F	F	T	T	F
F	F	T	T	F	F	T	T

Set Theory, Logic, Relations & Functions Notes

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2025/26

Course calendar – Block 1

Week	Date	Teaching Topics	pages
1	29/09/2025	Sets 1. Induction 2. 1.1- 1.5	7 – 12
2	06/10/2025	Sets: 1.6 – 1.9	17 – 20
3	13/10/2025	Sets: 1.10 – 1.12	23 - 28
4	20/10/2025	Logic (1): 2.1 – 2.4	31 – 43
5	27/10/2025	Logic (1): 2.5 – 2.8	43 - 49
6	03/11/2025	Logic (1): 2.9	50 - 52
7	10/11/2025	Logic (2): 2.10 – 2.15	56 – 65
8	17/11/2025	Relations	69 – 73
9	24/11/2025	Functions	74 - 87
10	01/12/2025	Recap, Revision & Past papers	
11	08/12/2025	Recap, Revision & Past papers	

Reading list:

1. K. H. Rosen, *Discrete Mathematics and Its Applications*, 8th Global Ed., McGraw-Hill, 2018.
2. Rowan Garnier and John Taylor , *Discrete Mathematics for New Technology*, University of Brighton, UK, Second Edition 2002.
3. Eccles, Peter J. (1997) [An introduction to mathematical reasoning: numbers, sets, and functions](#)
4. Lipschutz, Seymour, Lipson, Marc (2007), Schaum's outline of discrete mathematics.
5. Finan, Marcel B. (2001), Lecture Notes in Discrete Mathematics, Arkansas Tech. University, USA.
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CHAPTER 1 – SETS AND ALGEBRA OF SETS (1)

1.1 Introduction

Historical Note: Georg Cantor, a German mathematician, is the brainchild of set theory. His work was done in the 1800s. Some textbooks will use M to denote a generic set. This comes from the German word *Menge*.

A **set** is a well-defined collection of objects having a certain property. The objects belonging to a set are called **elements** or **members** of the set. A set has no duplicates and the order of the objects in the set is not important.

Sets are usually denoted by **capital letters** and set elements are denoted by lower case letters.

The symbol ' $\in$ ' is used to denote '**is an element of**' or '**belongs to**' or '**a member of**'.

Example: The collection of all students taking this unit is a well-defined set.
The elements of this set are the students taking the unit.

Other examples at the lecture

1.2 Description of sets and notations

The following axiom is really the heart of set theory:

The Axiom of Extension:

- a set is completely defined by its elements
- order of elements is irrelevant
- an element may be listed twice - also irrelevant

A set can be described in several ways:

1.2.1 Verbal form

Example

- i. The set of all natural numbers, say $\mathbb{N}$.
- ii. The set of all letters in the alphabet, say A .
- iii. The set of all real numbers, say $\mathbb{R}$.
- iv. The set of all integer numbers, say $\mathbb{Z}$.

1.2.2 List or tabular form

This is where all the elements of the set are listed individually, separated by commas and enclosed in curly brackets $\{..., ..., ...\}$.

Example

The sets N and A in an above example can be described by the following tabular form:

- i. $N = \{1, 2, 3, 4, \dots\}$
- ii. $A = \{a, b, c, d, \dots, z\}$

Example

Let D be the set of all positive integers (i.e. natural numbers, the set N) between 2 and 16 inclusive. Rewrite the set using the set-builder form.

$$D = \{x | x \in N, 2 \leq x \leq 16\}.$$

This is read as: D is the set of all elements x such that x belongs to N and $2 \leq x \leq 16$.

1.3. Definition of Numbers - Standard Sets

- **Natural numbers, N**

A natural number is a positive whole number:

$$N = \{1, 2, 3, 4, \dots\} \text{ or } N = \{x | x \text{ is a natural number}\}.$$

Note: Be aware that in some texts the set N also includes the number 0. When we refer to N we make this clear.

- **Whole numbers, W**

$$W = \{0, 1, 2, 3, 4, \dots\} = \text{set of whole numbers.}$$

- **Integer numbers, $\mathbb{Z}$**

An integer number is a positive or negative whole number including zero:

$$\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, 4, \dots\} \text{ or } \mathbb{Z} = \{x | x \text{ is an integer number}\}.$$

$$\mathbb{Z}^+ = N = \{1, 2, 3, 4, \dots\} = \text{set of positive integers.}$$

$$\mathbb{Z}^- = \{-1, -2, -3, -4, \dots\} = \text{set of negative integers.}$$

- **Rational numbers, $\mathbb{Q}$**

A rational number is a ratio of two integers in the form $\frac{p}{q}$ where $q \neq 0$.

$$\mathbb{Q} = \{x | x \text{ is a rational number}\} = \{x | x = \frac{p}{q} \text{ where } p \in \mathbb{Z}, q \in \mathbb{Z} \text{ and } q \neq 0\}$$

- **Irrational numbers, I**

These are numbers such as $\sqrt{2}$, $\sqrt[3]{5}$ and π which are not replaceable by the ratio of two integers:

$$I = \{x | x \text{ is an irrational number}\} = \{x | x \text{ is not rational number}\}.$$

- **Real Numbers, $\mathbb{R}$**

Real numbers are the collection of all rational and irrational numbers.

$$\mathbb{R} = \{x \mid -\infty < x < +\infty\}$$

$\mathbb{R}^+ = \{x \in \mathbb{R} \mid x > 0\}$ = set of all positive real numbers.

$\mathbb{R}^- = \{x \in \mathbb{R} \mid x < 0\}$ = set of all negative real numbers.

- **Complex Numbers, $\mathbb{C}$** $Z = a + ib$

Other examples will be discussed in the lecture

1.3.1 Intervals

Certain subsets of $\mathbb{R}$, called intervals, will prove useful in our discussions. They are defined below as:

Closed interval	$[a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\}$
Closed-open interval	$[a, b) = \{x \in \mathbb{R} \mid a \leq x < b\} = [a, b[$
Open-closed interval	$(a, b] = \{x \in \mathbb{R} \mid a < x \leq b\} =]a, b]$
Open interval	$(a, b) = \{x \in \mathbb{R} \mid a < x < b\} =]a, b[, \text{ where } a < b.$

A bracket at an endpoint indicates it is included in the set, whereas a parenthesis indicates it is not included.

The set $\{x \in \mathbb{R} \mid x \geq a\}$ is denoted by $[a, \infty)$ using the **infinity symbol** ∞ .

Likewise, the set $\{x \in \mathbb{R} \mid x \leq a\}$ is denoted by $(-\infty, a]$.

Example

$$\begin{aligned} x \in [4, 6] &= 4 \leq x \leq 6, & x \in [4, 6) &= 4 \leq x < 6 \\ x \in (4, 6] &= 4 < x \leq 6, & x \in (4, 6) &= 4 < x < 6 \end{aligned}$$

Other examples will be discussed in the lecture

1.4. Relations between Sets

1.4.1 Subsets

A is a **subset** of B if every element in A is also an element in B , we write $A \subseteq B$.

Formally, this means that every element x , if $x \in A$, then $x \in B$.

If A is not a subset of B , that is, if at least one element of A does not belong to B , we write

$$A \not\subseteq B$$

Example

Let $A = \{1, 2, 3\}$. Then by using the elements of set A the following subsets of A can be formed:
 $\{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}.$

Definition: Proper Subset:

If the set A is a subset of B and $A \neq B$, then A is a ‘**proper subset** of B ’ or ‘ A is properly contained in B ’. We will write $A \subset B$. For example: $\{1, 2\} \subset \{1, 2, 3\}$.

However, if the possibility of $A=B$ is not excluded from the collection of subsets, then we use the notation ' $\subseteq$ '. For example, $\{1,2,3\} \subseteq \{1,2,3\}$.

Note that of all the subsets of a given set A , only A itself is '**improper**' (i.e. it is not a proper subset of A).

The symbol of proper subset is ' $\subset$ ' which implies completely contained in and not same as but the symbol of improper subsets is ' $\subseteq$ ' which implies **contained or same as** and this context more to be as same or equal only. Even if the improper subset of a set is same as that the set but we never use the symbol equal to " $=$ " to represent it but the symbol " $\subseteq$ " is only used.

If there are ' n ' elements in a given set then there can be only one subset of that set which is exactly same as that given set. In total, we have 2^n subsets possible for a given set and since only one is an improper subset, therefore we get $2^n - 1$ proper subsets of a given set.

Example

- i. $P = \{x\}$, $T = \{x, y, z\}$ and $M = \{v, w, x, y, z\}$ are all subsets of $L = \{v, w, x, y, z\}$.
Therefore, we may write $P, T \text{ \& } M \text{ are } \subseteq L$, although the **precise** statements for proper subsets P and T are $P \subset L$, $T \subset L$ but $M \subseteq L$ is improper subset.
- ii. $S_1 = \{1,2,3\}$ and $S_2 = \{1,2\}$, then $S_2 \subset S_1$ but $S_1 \not\subset S_2$.
- iii. $N \subseteq Z$, conveys the fact that every natural number is also an integer.
- iv. $Z \subseteq Q$, conveys the fact that every integer number is also a rational number since, for example, $3 = \frac{3}{1}$.
- v. $Q \subseteq R$, conveys the fact that every rational number is also a real number.
- vi. Observe that $N \subseteq Z \subseteq Q \subseteq R \subseteq C$.

Other examples will be discussed in the lecture

1.4.2 Special sets

• The Universal set

The elements of all sets under consideration are said to belong to a set called 'the universal' set denoted by ' U '.

For example:

1. In human population studies, the universal set consists of all people in the world.
2. In plane geometry, the universal set consists of all points in the plane.
3. U can be considered to be the set of all real numbers R .
4. U can be considered to be the set of whole numbers 1 to 10 (i.e. $U = \{1, 2, 3, \dots, 9, 10\}$).

• The Null or Empty set

The set containing no elements is called an empty or null set and is denoted by $\{\}$ or ' $\emptyset$ '. For example, consider the empty set $\{x|x \in N \text{ and } x < 0\} = \emptyset$.

The following properties hold for a null set:

- i. $\emptyset$ is a subset of every set A .
- ii. $\emptyset$ is a proper subset of every set $A \neq \emptyset$.
- iii. For any set A , we have $\emptyset \subseteq A \subseteq U$

1.4.3 Set Equality

Two sets A and B are **equal** (denoted as $A=B$), if and only if they have exactly the same elements. That is each member of A is a member of B and vice versa. This is expressed formally by saying " $A \subseteq B$ and $B \subseteq A$."

Example

- i. $L = \{\text{Prolog, Lisp, Modula2, Java, C}\}$ and $N = \{\text{Lisp, Modula2, C, Prolog, Java}\}$
- ii. $\{3, 6, 9, 12, \dots\} = \{x \in \mathbb{Z} \text{ and } x = 3m \text{ with } 0 < m \in \mathbb{Z}\}$.
- iii. $A = \{x | x \text{ is a vowel of the alphabet}\}$ and $B = \{a, e, i, o, u\}$
- iv. $C = \{2, 3, 4\}$ and $D = \{x | x \text{ is a digit in the numeral } 23432\}$

Theorem: Properties of Set Equality

- (a) For any set A , $A = A$. (Reflexive Property)
- (b) If $A = B$, then $B = A$. (Symmetric Property)
- (c) If $A = B$ and $B = C$, then $A = C$. (Transitive Property)

Theorem:

- 1. For any set A , we have $\emptyset \subseteq A \subseteq U$.
- 2. For any set A , we have $A \subseteq A$.
- 3. If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.
- 4. $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$.

Other examples will be discussed in the lecture

1.4.4 The Complement set

The set of all the elements under discussion that are not in the set A is called the complement of A and is denoted by A' (or A^c). In set-builder form A^c can be expressed as $A^c = \{x|x \in U, x \notin A\}$.

Example

- i. Let $U = \mathbb{R}$ the set of all real numbers. Thus, if A is the set of negative real numbers

including zero (i.e. $A = \{x \mid -\infty < x \leq 0\}$), then A^C is the set of positive real numbers (i.e. $A^C = \{x \mid 0 < x < \infty\}$).

- ii. Let $U = \mathbb{Z}^+$ the set of all positive integer numbers. Thus, if A is the set of even integers including zero (i.e. $A = \{0, 2, 4, 6, \dots\}$), then A^C is the set of odd integers (i.e. $A^C = \{3, 5, 7, \dots\}$).
- iii. Let U be the set of students at a university, and suppose M and F denote, respectively, the male and female students in U . Then

$$M^C = F \quad \text{and} \quad F^C = M$$

1.4.5 The Relative complement set $A \setminus B$

The *relative complement* of a set B with respect to a set A or, simply, the *difference* of A and B , denoted by $A \setminus B$, is the set of elements which belong to A but which do not belong to B ; that is $A \setminus B = \{x \mid x \in A, x \notin B\}$
 The set $A \setminus B$ is read “ A minus B ” many texts denote $A \setminus B$ by $A - B$.

Example

Consider the sets $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6, 7\}$ and $C = \{6, 7, 8, 9\}$

Then

$$A \setminus B = \{1, 2\}, \quad B \setminus C = \{3, 4, 5\}, \quad B \setminus A = \{5, 6, 7\} \quad \text{and} \quad C \setminus B = \{8, 9\}.$$

1.4.6. The unit set

A set containing only **one** element is called a **unit** set.

For example $A = \{x\}$ is a unit set whose only member is x .

Note that $A = \{x\}$ is not the same as $A = x$.

Other examples will be discussed in the lecture

1.5. Venn diagrams

John Venn was born in Hull, England in 1834. In 1853 he went to Gonville and Caius College Cambridge and graduated in 1857 becoming fellow of the college. For the next 5 years he went into priesthood and returned to Cambridge in 1862 to teach logic and probability theory.

John Venn is popular known as the person who developed a graphically way to look at sets and this graph become known as a *Venn diagram*. The sets were represented by oval or circular shape figures but they can be any shape.

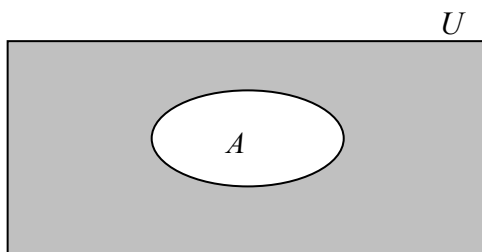
The properties and relations of sets can be displayed with Venn diagrams. The diagrams use a rectangle whose interior points are assumed to represent the elements of the universal set U :



U shaded

Any subset of the universal set (U) is represented by a circle (or a loop) drawn inside the rectangle U . Thus, any set A and its complement can be shown by Venn diagrams as:

What does the Venn diagram of A^c look like?

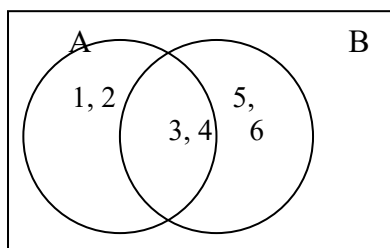


A^c (complement of A) is shaded

You may see A^c written as $\bar{A}$ or A' in other mathematical literature

Example

Use a Venn diagram to display the sets $A = \{1,2,3,4\}$ and $B = \{3,4,5,6\}$.



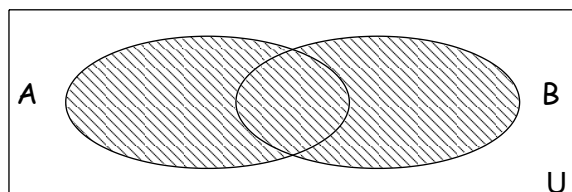
Other examples will be discussed in the lecture

See Venn Diagrams Power point

<http://www.math.tamu.edu/~kahlig/venn/venn.html>

1.6. Set Operations

1.6.1 The Union of two sets A and B is $\{x|x \in A \text{ or } x \in B\}$, denoted as $A \cup B$.

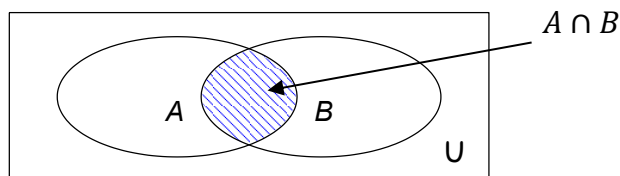


$A \cup B$ (A union B) is shaded

Let $A = \{2, 4, 5\}$ and $B = \{1, 4, 6, 8\}$.

Then, $A \cup B = \{1, 2, 4, 5, 6, 8\}$

1.6.2 The Intersection of two sets A and B is $\{x|x \in A \text{ and } x \in B\}$, denoted as $A \cap B$.



$A \cap B$ (A intersection B) is shaded

Examples:

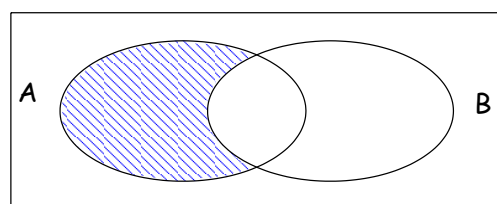
a) Let $A = \{2, 4, 5\}$ and $B = \{1, 4, 6, 8\}$ Then, $A \cap B = \{4\}$.

Note: A set that has only one element, such as $\{4\}$, is sometimes called a singleton set.

b) Let $A = \{2, 4, 5\}$ and $B = \{1, 3\}$. Then $A \cap B = \emptyset$.

1.6.3 The Difference of two sets A and B is the set of all the elements in A but not in B , $\{x|x \in A \text{ and } x \notin B\}$, denoted as $A \setminus B$ ($A - B$) or $A \sim B$

What does the Venn diagram of $A \setminus B$ look like?



$A \setminus B$ is shaded

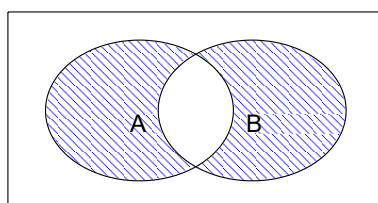
Examples:

- a. Let $B = \{2, 3, 6, 10, 13, 15\}$ and $A = \{2, 10, 15, 21, 22\}$.
Then $B \setminus A = B - A = \{3, 6, 13\}$.
- b. Let X be the set of natural numbers and A be the set of odd natural numbers. Then,
 $X - A =$ the set of even natural numbers; or $X - A = \{x \mid x \text{ is a natural number and } x \text{ is even}\}$.

1.6.4 The Symmetric difference of sets A and B

denoted by $A \oplus B$, consists of those elements which belongs to A or B but not to both. That is,
 $A \oplus B = (A \cup B) \setminus (A \cap B)$ or $A \oplus B = \{x \mid x \in A, x \in B, x \notin A \cap B\}$ or
 $A \oplus B = (A \setminus B) \cup (B \setminus A)$

What does the Venn diagram of this look like?



$A \oplus B$ is shaded

Example

Suppose $U = N = \{1, 2, 3, 4, \dots\}$ is the universal set.

Let $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6, 7\}$, $C = \{2, 3, 8, 9\}$, $E = \{2, 4, 6, 8, \dots\}$

Then:

$$A \oplus B = \{1, 2, 5, 6, 7\}, \quad B \oplus C = \{2, 4, 5, 6, 7, 8, 9\}$$

$$A \oplus C = \{1, 4, 8, 9\}, \quad A \oplus E = \{1, 3, 6, 8, 10, \dots\}$$

The operations $\cup, \cap$ and $\setminus$ can be repeated or combined to derive more sets.

Example

Let $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6, 7\}$ and $C = \{6, 7, 8\}$.

. Express in tabular form the composite sets:

- $(A \cup B) \cap C = \{1, 2, 3, 4, \neq 5, 6, 7\} \cap \{6, 7, 8\} = \{6, 7\}$.
- $(A \cup C) \setminus (B \cup C) = \{1, 2\}$

Note that in any mathematical expression containing brackets (), we must perform the operations inside the bracket(s) first

Other examples will be discussed in the lecture

1.6.5 The disjoint of two sets

A and B are said to be disjoint or nonintersecting if they have no elements in common or, using the definition of intersection, $A \cap B = \varphi$, the empty set.

Suppose $S = A \cup B$ and $A \cap B = \varphi$, then S is called the *disjoint union* of A and B.

Example

Let $A = \{1,2,3,4\}$, $B = \{3,4,5,6,7\}$ and $C = \{6,7,8\}$.

Then $A \cup B = \{1,2,3,4,5,6,7\}$,

$$A \cap B = \{3,4\},$$

$$C \setminus B = \{8\}.$$

Note that A and C are **disjoint** (i.e. their intersection is the empty set).

The following properties of union and intersection should be noted.

Property 1: Every element x in $A \cap B$ belongs to both A and B ; hence x belongs to A and x belongs to B . Thus $A \cap B$ is a subset of A and of B ; namely

$$A \cap B \subseteq A \quad \text{and} \quad A \cap B \subseteq B$$

Property 2: An element x belongs to the union $A \cup B$ if x belongs to A or x belongs to B . Hence every element in A belongs to $A \cup B$, and every element in B belongs to $A \cup B$.

That is, $A \subseteq A \cup B$ and $B \subseteq A \cup B$.

1.6.6 Duality:

Let E be an equation of set algebra. The ***dual*** E^* of E is the equation obtained by replacing each occurrence of $\cup, \cap, U, \emptyset$ in E by $\cap, \cup, \emptyset, U$ respectively.

Example: $(U \cap A) \cup (B \cap A) = A$ is $(\emptyset \cup A) \cap (B \cup A) = A$.

1.7 Finite and Countable Sets

Sets can be finite or infinite. A set is **finite** if it is an empty or if it consists of exactly n distinct elements where n is a positive integer; otherwise it is **infinite**.

Example 1

Let M be the set of the days of the week; that is,

$$M = \{\text{Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday}\}$$

Then M is **finite**.

Example 2

Let $P = \{x \mid x \text{ is a river on the earth}\}$.

Although it may difficult to count the number of rivers on the earth, P is a **finite** set.

Example 3

Let Y be the set of (positive) even integers, i.e. $Y = \{2, 4, 6, \dots\}$.

Then Y is an **infinite** set.

Example 4

Let I be the *unit interval* of real numbers, i.e. $I = \{x \mid 0 \leq x \leq 1\}$.

Then I is also an **infinite** set.

Example 5

The sets of Natural Numbers, N , Integer numbers, Z , Rational numbers, Q , Irrational numbers, I , and Real numbers, R , are **infinite** sets.

A set S is **countable** if it is **finite** or if its elements of S can be **arranged** in the form of a sequence, in which case S is said to be **countable infinite**; otherwise the set is **uncountable**.

The set in the example 3 is countable infinite, whereas it can be shown that the set in the example 4 is uncountable.

1.8. Set Cardinality

The cardinality of a finite set A is **the number** of elements that it contains. Cardinality is denoted by $n(A)$ or $|A|$ (i.e. the number of elements in a finite set A).

Remark: The term *cardinality* comes from the common usage of the term *cardinal number* as the size of a finite set.

Example: Let A be the set of odd positive integers less than 10. Then $n(A)=|A| = 5$.

Example: Let S be the set of letters in the English alphabet. Then $n(S)=|S| = 26$.

Example: Because the null set has no elements, it follows that $n(\emptyset)=|\emptyset| = 0$.

Note the following cardinality for the combination of sets:

If A and B are finite **disjoint** sets. Then $A \cup B$ is finite and

$$n(A \cup B) = n(A) + n(B).$$

If A be a subset of a finite universal set U , then U is the disjoint union of A and its complement A^c . Thus according to above

$$n(U) = n(A) + n(A^c)$$

If A be a subset of a finite universal set U . Then

$$n(A^c) = n(U) - n(A)$$

Example: if there are 20 male students in a class of 35 students,
then there is $35 - 20 = 15$ female students.

If A and B are finite sets. Then

$$n(A \setminus B) = n(A) - n(A \cap B)$$

Example: suppose an archery class A contains 35 students, and 15 of them are also in a bowling class B . Then

$$n(A \setminus B) = n(A) - n(A \cap B) = 35 - 15 = 20$$

(Inclusion-Exclusion Principal)

Suppose A and B are finite sets, not necessarily disjoint. Then $A \cap B$ and $A \cup B$ are finite, and

$$n(A \cup B) = n(A) + n(B) - n(A \cap B).$$

That is, we find the number of elements in A or B (or both) by first adding $n(A)$ and $n(B)$ (inclusion) and then subtracting $n(A \cap B)$ (exclusion) since the elements in $A \cap B$ were counted twice.

Suppose A , B and C are finite sets. Then $A \cup B \cup C$ is finite and

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C).$$

Example

Suppose an art class A has 25 students and 10 of them are taking a biology class B . Then the number of students in class A which are not in class B is:

$$n(A \setminus B) = n(A) - n(A \cap B) = 25 - 10 = 15$$

Other examples will be discussed in the lecture

1.9 Classes of Sets, Power Sets

1.9.1. Classes of Sets: Sets can be elements of other sets. For example, each line in a set of lines is a set of points. We usually use the word *class* or *family* for such a set.

Example: The members of the class $\{\{2, 3\}, \{2\}, \{5, 6\}\}$ are the sets $\{2, 3\}$, $\{2\}$ and $\{5, 6\}$.

Example: Answer each of the following questions about sets:

1. Is $\{1, 2\} \subseteq \{\{1, 2\}, \{1, 3, 4\}\}$?
2. Is $\{1, 2\} \in \{\{1, 2\}, \{1, 3, 4\}\}$?
3. What is $n(\{\{1, 2\}, \{2, 4\}, \{1, 3, 4\}\})$?

Solution:

1. No. The set $\{1, 2\}$ is an element of the set $\{\{1, 2\}, \{1, 3, 4\}\}$ but not a subset.
2. Yes. As mentioned above, the set $\{1, 2\}$ is an element of the set $\{\{1, 2\}, \{1, 3, 4\}\}$.
3. $n(\{\{1, 2\}, \{2, 4\}, \{1, 3, 4\}\}) = 3$

1.9.2. Power Sets:

The set of all subsets of a given set A is called the '*Power*' set of A and is denoted by $P(A)$ (i.e. $P(A) = \{x | x \subseteq A\}$).

Example: Let $A = \{a, b, c\}$, then $P(A) = \{\emptyset, A, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}\}$

In general, if A is a finite set and has k elements, then $P(A)$ will have 2^k elements. Or in other words, for any finite set A , if $n(A) = k$, then $n(P(A)) = 2^k$. Or the power set of any finite set A has **cardinality** 2^k , where k is the cardinality of set A .

Example: from the above example, $n(P(A)) = 2^3 = 8 = \text{Cardinality}$.

Example:

- i. Let $A = \{1, 2, 3\}$ then

$$P(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\} \text{ and } n(P(A)) = 2^3 = 8$$

- ii. Let $B = \{h, k\}$ then

$$P(B) = \{\emptyset, \{h\}, \{k\}, \{h, k\}\} \text{ and } n(P(B)) = 2^2 = 4.$$

Note that the empty set and the set itself are members of this set of subsets.

Example: What is the power set of the empty set? What is the power set of $\{\emptyset\}$?

Solution: The empty set has exactly one subset, namely, itself. Consequently, $P(\emptyset) = 2^0 = 1 = \{\emptyset\}$.

The set $\{\emptyset\}$ has exactly two subsets, namely, $\emptyset$ and the set $\{\emptyset\}$ itself. Therefore $P(\{\emptyset\}) = 2^1 = 2 = \{\emptyset, \{\emptyset\}\}$.

1.9.3 Partition

A partition of a set A is a subdivision of A into **nonempty** subsets which are **disjoint** and whose **union is A** , i.e. is a class of nonempty subsets of A such that each $a \in A$ belongs to a unique subset.
The subsets in a partition are called *cells*.

More generally, Let I be an index set and P a family of subsets S_i of a non-empty set S , where $i \in I$ (i.e. $P = \{S_1, S_2, S_3, S_4, \dots, S_i, \dots\}$). Then P is a **partition** of S if:

- Each set $S_i \neq \varnothing$
- The subsets are pairwise disjoint; that is, $S_i \cap S_j = \varnothing$ if $i \neq j$.
- The union of the subsets S_i is S ; that is, $\bigcup_{i \in I} S_i = S$.

Example: Determine which of the following classes of subsets of $A = \{1, 2, 3, \dots, 8, 9\}$ is a partition of A :

- i) $[\{1, 3, 5\}, \{2, 6\}, \{4, 8, 9\}]$
- ii) $[\{1, 3, 5\}, \{2, 4, 6, 8\}, \{5, 7, 9\}]$
- iii) $[\{1, 3, 5\}, \{2, 4, 6, 9\}, \{7, 8\}]$

Solution:

- i) Is not a partition of A since $7 \in A$ but 7 does not belong to any of the cells.
- ii) Is not a partition of A since $5 \in A$ and 5 belongs to $\{1, 3, 5\}$ and $\{5, 7, 9\}$.
- iii) Is partition of A since each element of A belongs to exactly one cell.

Example: Find all partitions of $V = \{1, 2, 3\}$.

Other examples will be discussed in the lecture

EXERCISE 1 – SETS AND ALGEBRA OF SETS (1)

1. Express the following sets in an alternative tabular form.
 - (a) $Z_7 = \{x | 1 \leq x \leq 7\}$
 - (b) $A = \{x | x = 1 + \frac{1}{y}, y \in Z_7\}$.
 - (c) $B = \{x | x \in N \text{ and } \frac{24}{x} \in N\}$.
2. List the elements of each of these sets:
 - a) $A = \{x | x \in N \text{ and } x > 6 \text{ and } x < 10\}$
 - b) $B = \{x | x \in N \text{ and } x > 7 \text{ and } x \leq 8\}$
 - c) $C = \{x | x \in N \text{ and } x < 20 \text{ and } x \text{ is a prime number}\}$Write down the cardinality of each set.
3. In this problem the universal set is
$$U = \{x | x \in N \text{ and } x \leq 12\}$$
Where,
$$A = \{1, 2, 3, 4, 5, 6, 7\} \text{ and } B = \{4, 5, 6, 7, 8, 9, 10\}.$$
List the subsets of:
 $a) A'; b) B'; c) A \cap B; d) A \cup B; e) A \setminus B; f) A' \cup B'; g) A \cup B'; h) B' \setminus A; i) A' \cap B'$
4. Draw Venn diagrams for the following sets:
 - (a) $A = \{1, 3, 5, 7\}, B = \{1, 2, 3, 4\}, C = \{1, 2, 5, 6\}$.
 - (b) $A = \{1, 2, 3, 4\}, B = \{2, 4\}, C = \{2\}$.
5. Which of the following sets are equal or which is a subset of the other? Justify your answer.
 - (a) $A = \{x | x = \frac{1}{n}, 1 < n \in N\}, B = \{\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}\}$.
 - (b) $A = \{a, b, c\}, B = \{a, a, b, b, c, c\}$.
 - (c) $A = \{m | m \in N \text{ and } \frac{24}{m} \in N\}, B = \{m | m \in N \text{ and } \frac{48}{m} \in N\}$.
6. Let N_e represent the set of all even natural numbers. If the universal set $U = \{1, 2, 3, 4, 5, 6\}$ and set $A = \{x | x \in N_e\}$, describe in tabular form the following sets:
 - (a) A^c
 - (b) $B = \{y | y = x^3 \text{ and } x \in A\}$
 - (c) $P(A)$
 - (d) $P(A^c)$.
7. In a survey of 36 PC users it was found that 11 used Word, 14 used Excel and 16 used Access. 8 used none of these packages. 4 used Word and Excel, 5 used Word and Access and 3 used all three packages.
 - a. How many used Excel or Access (or both)?
 - b. How many used both Excel and Access?
 - c. How many used only Access?

8. Amongst 25 PC users 10 used Word, 12 Excel and 15 Access. 3 used none of these packages. 6 used Word and Access, 5 used Excel and Access and 3 used all three packages.

- How many used Word and Excel but not Access?
- How many used only Word?
- How many used only Excel?

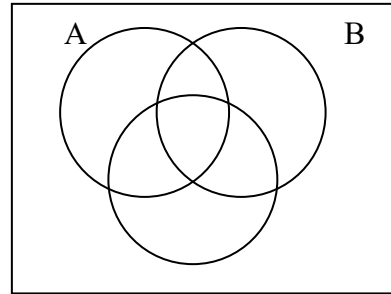
9. Find the cardinality of the following power sets:

$$P(\emptyset), P(\{a\}), P(\{a, b\}), P(\{a, b, c\}), P(\{a, b, c, d\}).$$

Use your answers to verify the result that the power set A has cardinality 2^n , where n is the cardinality of set A.

10. The most general Venn diagram with three sets A, B, and C is shown below. Use copies of it to shade the following sets:

- $A \cup (B \cap C)$
- $(A \cup B) \cap (B \cup C)$
- $(A \cap B) \cup (A \cap C)$
- $A^c \cap (B \cup C^c)$
- $A \cup (B \setminus C)$
- $(A \cup B) \setminus C$.



DISCRETE MATHEMATICS

CHAPTER 1 – SETS AND ALGEBRA OF SETS (2)

1.10. Laws of Set Algebra (Set Identities)

The following properties and relationships hold for set operations.

SET ALGEBRA (Identity)	NAME
a. $A \cup A^c = U$ b. $A \cap A^c = \emptyset$	Complement laws
a. $A \cap U = A$ b. $A \cup \emptyset = A$	Identity laws
a. $A \cup U = U$ b. $A \cap \emptyset = \emptyset$	Domination laws
a. $A \cup A = A$ b. $A \cap A = A$	Idempotent laws
a. $(A^c)^c = A$	Double complement law
a. $A \cup B = B \cup A$ b. $A \cap B = B \cap A$	Commutative laws
a. $A \cup (A \cap B) = A$ b. $A \cap (A \cup B) = A$	Absorption laws
a. $(A \cup B) \cup C = A \cup (B \cup C) = A \cup B \cup C$ b. $(A \cap B) \cap C = A \cap (B \cap C) = A \cap B \cap C$	Associative laws
a. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ b. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ c. $A - (B \cup C) = (A - B) \cap (A - C)$ d. $A - (B \cap C) = (A - B) \cup (A - C)$	Distributive laws
a. $(A \cup B)^c = A^c \cap B^c$ b. $(A \cap B)^c = A^c \cup B^c$	De Morgan's laws

Theorem:

Each of the following conditions is equivalent to $A \subset B$:

1. $A \cap B = A$ 2. $A \cup B = B$ 3. $B^c \subset A^c$

4. $A \cap B^c = \emptyset$ 5. $B \cup A^c = U$

Example1:

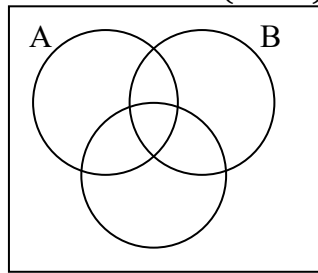
Use the relations of set algebra to prove that:

- i. $A \cap (A^c \cup B) = A \cap B$.
 $A \cap (A^c \cup B) = (A \cap A^c) \cup (A \cap B)$ Distributive
 $= (\emptyset) \cup (A \cap B)$ Exclusion
 $= (A \cap B)$ Identity.
- ii. $A \cup (A \cap B) = A$.
 $A \cup (A \cap B) = (A \cap U) \cup (A \cap B)$ Identity
 $= A \cap (U \cup B)$ Distributive
 $= A \cap U$ Domination
 $= A$ Identity

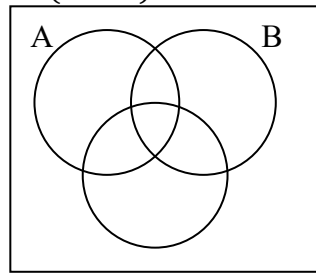
Example2:

Draw a Venn diagram to illustrate the following equivalences: (Diagrams to be completed in the class)

- a. The associative law – $(A \cup B) \cup C = A \cup (B \cup C)$.

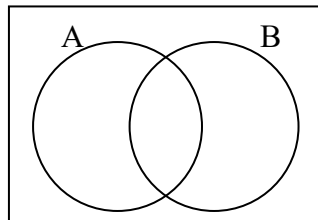


$(A \cup B) \cup C$

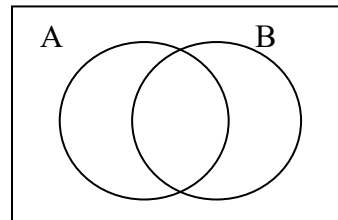


$A \cup (B \cup C)$

- b. De Morgan's law – $(A \cup B)^c = A^c \cap B^c$.



$(A \cup B)^c$



$A^c \cap B^c$

Other examples will be discussed in the lecture

1.11 Cartesian or Cross products

So far, in discussing sets, the order of the elements of a set has been irrelevant. Here, we introduce the concept of ordered sets, where the order in which the elements are listed is important.

In Mathematics the notation (a, b) is used to represent an ordered pair. That is the set of two elements $\{a, b\}$ together with the condition that element 'a' is regarded as the first.

Definition: -

Let A and B be any two sets. The set of all ordered pairs such that the first member of the ordered pair is an element of A and the second member is an element of B , is called the **Cartesian product** of A and B written as $A \times B$ (which reads as A cross B), i.e.

$$A \times B = \{(x, y) | (x \in A) \text{ and } (y \in B)\}.$$

Similarly, the Cartesian product of 3 sets A, B and C can be expressed as:

$$A \times B \times C = \{(x, y, z) | (x \in A) \text{ and } (y \in B) \text{ and } (z \in C)\}.$$

For any n sets $A_1, A_2, A_3, \dots, A_n$ the set of all ordered n -tuples $\{a_1, a_2, a_3, \dots, a_n\}$ where $a_i \in A_i$ for $i = 1, 2, 3, \dots, n$ is called the Cartesian product of sets $A_1, A_2, A_3, \dots, A_n$ and is denoted by

$$A_1 \times A_2 \times \dots \times A_n = \{(a_1, a_2, a_3, \dots, a_n) | a_i \in A_i, i = 1, 2, 3, \dots, n\}.$$

Example:

- i. If $A = \{0, 1, 2\}$ and $B = \{2, 3\}$ are two sets then the Cartesian product of A and B is given as:

$$A \times B = \{(0, 2), (0, 3), (1, 2), (1, 3), (2, 2), (2, 3)\}.$$

- ii. If $A = \{1, 2\}$ and $B = \{p, q, r\}$, then the Cartesian product of A and B is given as

$$A \times B = \{(1, p), (1, q), (1, r), (2, p), (2, q), (2, r)\}.$$

- iii. If $A = \{1, 2\}$, $B = \{m, n\}$ and $C = \{p, q\}$, then the Cartesian product of A, B and C is given as:

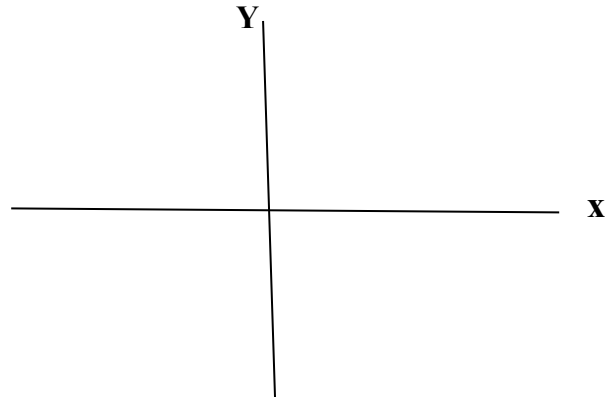
$$A \times B \times C = \{(1, m, p), (1, m, q), (1, n, p), (1, n, q), (2, m, p), (2, m, q), (2, n, p), (2, n, q)\}.$$

■

An important example of a Cartesian product is given by the set of all Cartesian coordinates in a plane. This product is given as $R \times R$, or R^2 .

1.12 Representation of Cartesian Products

- The Graphical Representation



- Table Representation

For the tabular representation of the cross product of two sets A and B, we need to draw a table or a 'rectangular array' in which the row number represents the elements of A and column number represents the elements of B, as shown in the following example.

Example:

Let $A = \{a, b\}$ and $B = \{w, x, y, z\}$. Using the tabular representation find the cross products:

i. $A \times B$

	w	x	y	z
a	(a, w)	(a, x)	(a, y)	(a, z)
b	(b, w)	(b, x)	(b, y)	(b, z)

(i.e. $A \times B = \{(a, w), (a, x), (a, y), (a, z), (b, w), (b, x), (b, y), (b, z)\}$)

ii. $B \times A$

	a	b
w	(w, a)	(w, b)
x	(x, a)	(x, b)
y	(y, a)	(y, b)
z	(z, a)	(z, b)

(i.e. $B \times A = \{(w, a), (w, b), (x, a), (x, b), (y, a), (y, b), (z, a), (z, b)\}$).

Note the differences between the elements of the cross products $A \times B$ and $B \times A$.

iii. $A \times A$ (to be completed by students).

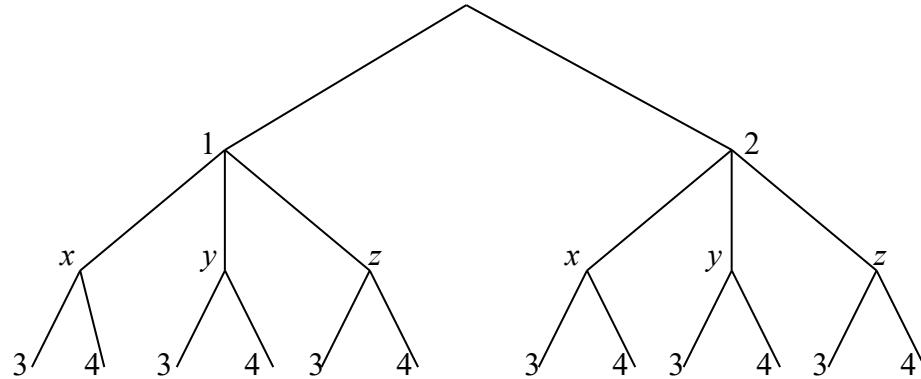
iv. $B \times B$ (to be completed by students).

- **The Tree Diagram Representation**

This is illustrated by the following examples.

Example:

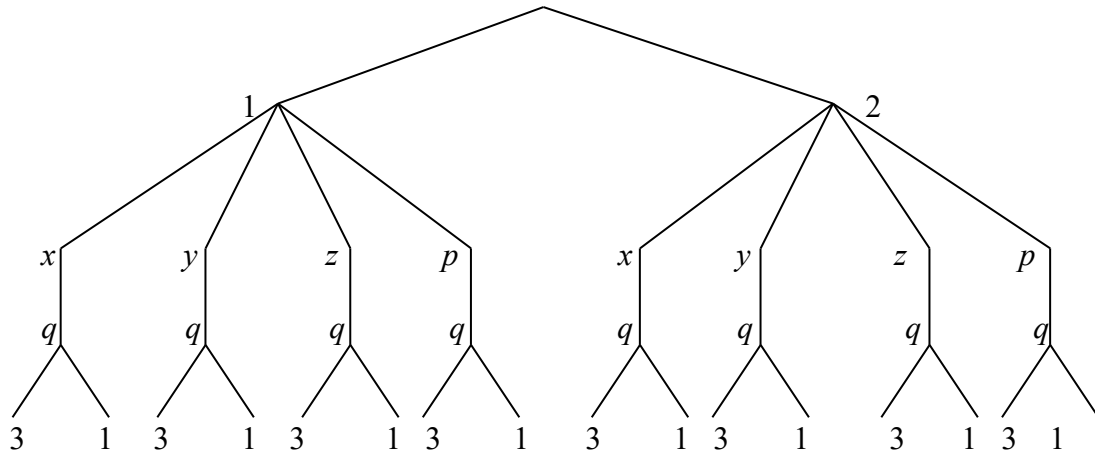
- i. Let $A = \{1,2\}$, $B = \{x, y, z\}$ and $C = \{3,4\}$. Find the Cartesian products of A , B and C using a tree representation.



Therefore, there are 12 ordered triplets of $A \times B \times C$, i.e.

$$A \times B \times C = \{(1, x, 3), (1, x, 4), (1, y, 3), (1, y, 4), (1, z, 3), (1, z, 4), (2, x, 3), (2, x, 4), (2, y, 3), (2, y, 4), (2, z, 3), (2, z, 4)\}.$$

- ii. Let $A = \{1,2\}$, $B = \{x, y, z, p\}$, $C = \{q\}$ and $D = \{3,1\}$. Find the Cartesian product $A \times B \times C \times D$, using a tree representation.



Therefore, there are 16 ordered quartets of $A \times B \times C \times D$, i.e.

$\{(1, x, q, 3), (1, x, q, 1), (1, y, q, 3), (1, y, q, 1), (1, z, q, 3), (1, z, q, 1), (1, p, q, 3), (1, p, q, 1)$
 $(2, x, q, 3), (2, x, q, 1), (2, y, q, 3), (2, y, q, 1), (2, z, q, 3), (2, z, q, 1), (2, p, q, 3), (2, p, q, 1)\}.$

There are two things worth noting :

1. $A \times B \neq B \times A$.

The Cartesian product deals with ordered pairs, so naturally the order in which the sets are considered is important.

2. $n(A \times B) = n(A).n(B)$.

The number of elements in the Cartesian products of sets can be found using the corresponding set cardinalities,

i.e. $n(A) = 2, n(B) = 4, n(C) = 1, n(D) = 2$, and
 $n(A \times B \times C \times D) = 16$.

Other examples will be discussed in the lecture

EXERCISE 2 – SETS AND ALGEBRA OF SETS (2)

1. If $U = \{2,4,6,8,10,12,14\}$, $A = \{2,4,6\}$, $B = \{2,6,10,14\}$ and $C = \{6,10,14\}$, find the following sets in tabular form:

a) A^C	(b) B^C	(c) C^C	(d) $A \cup B$	(e) $A \cup C$
(f) $B \cup C$	(g) $A - B$	(h) $A \cap B$	(i) $A \cap C$	(j) $B \cap C$
(k) $(A^C)^C$	(l) $(B \cap C)^C$	(m) $(A \cup B)^C$	(n) $(A \cup B) \cup C$	
(o) $(A \cup C) \cap (A \cap C)$	(p) $(A \cap B) \cap C$			
2. Find the union, the intersection, and the set difference of A and B, where $A = \{1,3,4,5\}$ and $B = \{3,5,7,8\}$.
3. Draw a Venn diagram for each of the sets:
(a) B^C (b) $(A \cup B)^C$
(c) $B - (A)^C$ (d) $A^C \cup B$.
4. Find the sets A and B where $A \cup B = \{1,2,3,4,5,6\}$, $A \cap B = \{2,4\}$, $A - B = \{1,6\}$. If the universe set is $U = \{1,2, \dots, 10\}$, find $(A \cup B)^C$ and $(A \cap B)^C$.
5. Given sets $A = \{3,4,5\}$ and $B = \{1,2,3,4\}$, use a tree diagram to find the Cartesian products (a) to (d):
(a) A^2
(b) $B \times A$
(c) B^2
(d) $A \times B$
(e) Find the set $(A \times A) \cap (B \times B)$
(f) Find the set $(A \times B) \cap (B \times A)$.
6. Using the properties of set algebra, prove the following:
(a) $(\emptyset \cup A) \cap (B \cup A) = A$
(b) $(A \cup B) \cap (A \cup B^C) = A$
(c) $A \cap (A \cup B) = A$
(d) $(U \cap A) \cup (B \cap A) = A$
(e) $(A \cap B) \cup (A \cap B^C) = A$
(f) $(A \cap B^C) \cup B = A \cup B$
(g) $(A \cap B^C) \cap A^C = \emptyset$
(h) $(A \cap B) \cup C = (A \cup C) \cap (B \cup C)$.
7. Rewrite the following statements using set notations:
(a) x is an element of X .
(b) a does not belong to A .
(c) set B is a subset of F .
(d) C is an empty set.

8. Let $A = \{a, b, c\}$. Which of the following statements are correct? Justify your answer for the incorrect ones.
- (a) $a \subset A$ (b) $a \in A$
(c) $\{a\} \subset A$ (d) $\{a\} \notin A$.
9. Let $X = \{x \mid x^2 = 4, x > 3\}$, Which of the following statements are correct? Justify your answer for the incorrect ones.
- (a) $X = 0$ (b) $X = \{0\}$
(c) $X = \emptyset$ (d) $X = \{\emptyset\}$.

DISCRETE MATHEMATICS

CHAPTER 2 – LOGIC (1)

2.1. Propositional Logic

Generally, logic deals with the consistency of statements. Any sentence or a statement that can be assigned the value **true (T)** or the value **false (F)**, but not both, is **a proposition**.

Example 1

The following are all propositions:

- i. $2 > 1$.
- ii. $\{1,2,3\} \cap \{3,4,5\} = \{3\}$.
- iii. The sum of any pair of even numbers is divisible by 3.

Propositions are commonly denoted by the letters p , q and r which are referred to as propositional variables. Propositional variables can only assume (or be assigned) two values true or false. There are also two propositional constants, T and F , that represent true and false. Therefore, any propositional variable can be assigned the value T or F . Propositional variables and propositional constants are **atomic** or simple propositions; that is, they cannot be further divided (i.e Cannot be expressed in terms of simpler propositions).

Example 2

The following are all propositions:

- p : The sun is shining.
- q : It is hot.
- r : Manchester is in the Northwest of England.

Example 3

The following are NOT propositions:

- i. What time is it?
- ii. Read this carefully.
- iii. $x+2=4$
- iv. $x+y=z$

Sentences *i* and *ii* are not propositions because they are not declarative sentences. Sentences *iii* and *iv* are not propositions because they are neither true or false. Note that each of sentences *iii* and *iv* can be turned into a proposition if we assign values to the variables.

2.2 Logical Connectives

We now turn our attention to methods for producing new propositions from those that we **Links** already have. These methods were discussed by the English mathematician George Boole in 1854 in his book *The Laws of Thought*. Many mathematical statements are constructed by combining one or more propositions. New propositions, called **compound propositions**, are formed from existing propositions using **logical operators**.

Definition: Compound Propositions

A compound proposition is formed from combining two or more simple propositions.

Therefore, all *non-atomic* propositions are compound propositions.

Consider the compound proposition:

‘I am a student at university, and I am studying a computing course’.

This compound proposition is constructed from two atomic propositions, ‘I am a student at university’ **and** ‘I am studying a computing course’. The word **and** which connects the two propositions is called a **logical connective**. All compound propositions contain at least one logical connective.

The logical value of a compound proposition depends upon the truth values of its component atomic propositions.

Example 4

Consider the compound proposition:

I have a good job **AND** I am happy.

This is true only if the simple proposition components: I have a good job and I am happy are both true.

We say that the compound proposition consisting of two simple propositions joined together by the connective AND is **true only when both component propositions are true**.

Finding the truth value of a more complex proposition is more difficult. A **truth table** can be used, for all cases, to give the truth values of the proposition under all possible assignments.

There are three basic logical connectives: **AND**, **OR** and **NOT**.

2.2.1 The Connective AND – Conjunction

Let p and q be propositions. The compound of two propositions p and q connected by **AND** is called the conjunction of two propositions ($p \wedge q$). In a symbolic description of a compound proposition the symbol ' $\wedge$ ' is used instead of the word AND .

The truth table for the connective AND ($\wedge$) for the simplest compound proposition of two simple propositions p and q is given by

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

We say that the compound proposition consisting of two simple propositions joined together by the connective AND is **true only when both component propositions p and q are true and is false otherwise.**

Note: in logic the word “but” sometimes is used instead of “and” in a conjunction.

Example 5

Let p and q represent the propositions,

p : Manchester is in the Northwest

q : Manchester is a city

$p \wedge q$ then translates as ‘Manchester is a city in the Northwest’.

Other examples will be discussed in the lecture

2.2.2 The Connective OR – Disjunction

Let p and q be propositions. The compound of two propositions p and q connected by **OR** is called the disjunction of two propositions ($p \vee q$). In a symbolic description of a compound proposition the symbol ' $\vee$ ' is used instead of the word OR.

The truth table for the connective OR for the simplest compound proposition of two simple propositions p and q is given by

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

**Thus, $p \vee q$ is false only if both p and q are false, and $p \vee q$ is true if p or q or both are true.
(i.e. the disjunction of two propositions is true if at least one sub proposition is true).**

Example 6

Let p and q represent the propositions,

p : There is a bug in the program.

q : There is an error in the input data.

Then, $p \vee q$ translates as 'There is a bug in the program or there is an error in the input data'.

Other examples will be discussed in the lecture

2.2.3 The Connective NOT– Negation

The negation of a proposition results in a proposition that has the opposite truth value to that of the original proposition. The symbol ‘ $\neg$ ’ is used for the negation and we write the negation of proposition p as $\neg p$. The connective $\neg$ may be translated into ‘It is not the case that’ or simply by the word ‘not’.

The truth table for the connective **NOT** is:

p	$\neg p$
T	F
F	T

Thus, if a proposition is true then its negation is false and if the proposition is false then its negation is true.

Remark: The notation for the negation operator is not standardized. Although $\neg p$ and $p^{\sim}$ are the most common notations used in mathematics to express the negation of p , other notations you might see are $\sim p$, $-p$, p' , Np , and $!p$.

Example 7

Let p represent the proposition,
 p : There is a bug in the program.

Then, $\neg p$ stands for the proposition ‘It is not the case that there is a bug in the program’ or ‘There is not a bug in the program’.

Example 8

Find the negation of the proposition “Michael’s PC runs Linux” and express this in simple English.

Solution: The negation is “It is not the case that Michael’s PC runs Linux.”
This negation can be more simply expressed as “Michael’s PC does not run Linux.”

Other examples will be discussed in the lecture

Remarks: the logical notation for the connectives “and”, “or”, and “not” are not completely standard. For example, some texts use:

$p \& q, p \cdot q$ or pq for $p \wedge q$

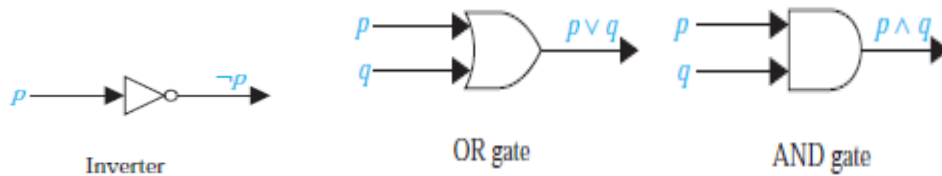
$p + q$ for $p \vee q$

$p' \text{ or } \sim p$ for $\neg p$

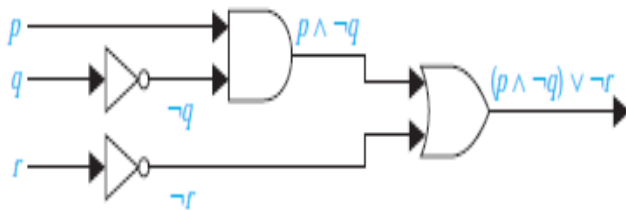
Basic Logic Circuits: Propositional logic can be applied to the design of computer hardware.

A **logic circuit** (or **digital circuit**) receives input signals $p_1, p_2, p_3, \dots, p_n$, each a bit [either 0 (off) or 1 (on)], and produces output signals $s_1, s_2, s_3, \dots, s_n$ each a bit.

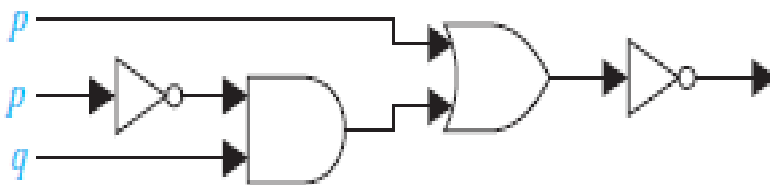
Complicated digital circuits can be constructed from three basic circuits, called gates, shown in Figure below.



We use combinations of these three basic gates to build more complicated circuits, such as that shown in Figure of the following examples,



Example: Determine the output for the combinatorial circuits in the figure below.



Solution: The output is $\neg(p \vee (\neg p \wedge q))$

In addition to the three basic logical connectives AND, OR and NOT, there are two implication connectives, **Conditional and Biconditional**.

2.2.4 Conditional or Implication proposition

In logical arguments the ‘**If ... then**’ construct or relationship is very often used. Conditional constructions are particularly used in programming for loop control and in branching.

For two propositions p and q , the conditional if p then q symbolically is denoted as $p \Rightarrow q$ or $p \rightarrow q$ and is frequently read “ p implies q ” or “ p only if q ”.

In the conditional statement $p \rightarrow q$, p is called the *hypothesis* (or *antecedent* or *premise*) and q is called the *conclusion* (or *consequence*).

The truth table for the conditional (or implication) of two simple propositions p and q is given by

p	q	$p \Rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Thus, $p \Rightarrow q$ is false only if p is true and q is false, otherwise $p \Rightarrow q$ is true.

Example 9

Let p and q represent the propositions,

p : The demand rises.

q : The companies expand.

Then, $p \Rightarrow q$ translates as ‘If demand rises, then companies expand’.

Some Phrasings of the Conditional

Each of the following is equivalent to the conditional $p \Rightarrow q$.

If p , then q .

p implies q .

q follows from p .

Not p unless q .

q if p .

p only if q .

Whenever p , q .

q whenever p .

p is sufficient for q .

q is necessary for p .

p is a sufficient condition for q .

q is a necessary condition for p .

When using the conditional proposition, $p \Rightarrow q$, various notations are used to refer to p and q :

p	$\Rightarrow$	q
antecedent		consequent
premise		conclusion
hypothesis sufficient condition		necessary condition.

The implication proposition $p \Rightarrow q$ has three associated related implications:

2.2.4.1.Converse:

If p and q are propositions, then the **converse** of $p \Rightarrow q$ is $q \Rightarrow p$, read as ‘if q , then p ’.

p	q	$p \Rightarrow q$	$q \Rightarrow p$
T	T	T	T
T	F	F	T
F	T	T	F
F	F	T	T

Example10

1. Give an example of a true implication whose converse is false.
2. Give an example of a true implication whose converse is also true.

Solution:

1.The implication “For all integers a and b , if a and b are odd, then $a + b$ is *even*” true.
Its converse “For all integers a and b , if $a + b$ is even, then a and b are *odd*” is false.

2. “If n is even, then n^2 is *even*” is true and has a true converse.

2.2.4.2 Inverse:

If p and q are propositions, then the **inverse** of $p \Rightarrow q$ is $\neg p \Rightarrow \neg q$ read as ‘if not p , then not q ’.

p	q	$p \Rightarrow q$	$\neg p \Rightarrow \neg q$
T	T	T	T
T	F	F	T
F	T	T	F
F	F	T	T

2.2.4.3 Contrapositive:

If p and q are propositions, then the **contrapositive** of $p \Rightarrow q$ is $\neg q \Rightarrow \neg p$
read as ‘if not q , then not p ’.

p	q	$p \Rightarrow q$	$\neg q \Rightarrow \neg p$
T	T	T	T
T	F	F	F
F	T	T	T
F	F	T	T

The **contrapositive** of “If $4 < 3$, then the moon is made of cheese” is “If the moon is not made of cheese, then $4 \geq 3$ ”.

Example 11

Consider the proposition:

$p \Rightarrow q$ If I study hard I get rich.

The converse is, $q \Rightarrow p$; if I get rich then I study hard.

The inverse is, $\neg p \Rightarrow \neg q$; if I don't study hard then I don't get rich.

The contrapositive $\neg q \Rightarrow \neg p$; if I don't get rich then I don't study hard.

Other examples will be discussed in the lecture

Observe the truth tables of $\neg p \vee q$ and $p \Rightarrow q$ are identical.

p	q	$\neg p$	$\neg p \vee q$
T	T	F	T
T	F	F	F
F	T	T	T
F	F	T	T

Accordingly, $p \Rightarrow q \equiv \neg p \vee q$

In general :-

1. *An implication and its contrapositive are logically equivalent.*
2. *The converse and inverse of an implication are logically equivalent.*
3. *An implication is not logically equivalent to its converse (and thus not to its inverse).*

We will prove the first statement by comparing the truth tables of $p \Rightarrow q$ and $\neg q \Rightarrow \neg p$.

p	q	$p \Rightarrow q$	$\neg q$	$\neg p$	$\neg q \Rightarrow \neg p$
T	T	T	F	F	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

2.2.5 Biconditional or Equivalence

For two propositions p and q , the compound proposition ' p **if and only if** q , or p **iff** q ' is referred to as the biconditional statement and symbolically is denoted as $p \Leftrightarrow q$.

The truth table for the biconditional or equivalence of two propositions p and q is given by:

p	q	$p \Leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

We can think of " $p \Leftrightarrow q$ " as follows:

1. p if and only if q .
2. For p to be true, it is necessary and sufficient for q to be true.
3. p implies q and q implies p .
4. p is equivalent to q .

A biconditional proposition " p if and only if q " involves two conditional propositions, " $If\ p, then\ q$ " and " $If\ q, then\ p$ ".

Thus, $p \Leftrightarrow q$ is true whenever p and q have the same truth values.

Note that $p \Leftrightarrow q$ is equivalent to $(p \Rightarrow q) \wedge (q \Rightarrow p)$; This can be shown by using truth tables.

Example 12

Let p and q represent the propositions,

p : x is even.

q : x is divisible by 2.

Then, then $p \Leftrightarrow q$ translates as ' x is even if and only if x is divisible by 2'.

Note that except for the connective $\neg$, which is the *unary connective*; that is, $\neg$ only negates a single proposition. All other connectives are *binary connectives*, which means that they require two propositions. The binary connectives $\wedge$, $\vee$ and $\Leftrightarrow$ are symmetric in the sense that the order of the two propositions joined by the connective does not affect the truth value of the resulting expression. That is the truth value of $p \wedge q$ is the same as the truth value of $q \wedge p$. The connective $\Rightarrow$ is not symmetric: $p \Rightarrow q$ and $q \Rightarrow p$ have different truth values.

The truth table of all connectives are summarized in the following table:

p	q	$\neg p$	$\neg q$	$p \wedge q$	$p \vee q$	$p \Rightarrow q$	$p \Leftrightarrow q$
T	T	F	F	T	T	T	T
T	F	F	T	F	T	F	F
F	T	T	F	F	T	T	F
F	F	T	T	F	F	T	T

Other examples will be discussed in the lecture

2.3. Tautology and Contradiction

2.3.1. Tautology:

A compound proposition that is **true in all possible** interpretations (or the logical values) of its components, is called a **tautology**.

For example, the proposition “ p or not p ” that is, $p \vee \neg p$, is a tautology

p	$\neg p$	$(p \vee \neg p)$
T	F	T
F	T	T

Example 1

Consider the truth table of the compound proposition, $(p \Rightarrow q) \vee (q \Rightarrow p)$.

p	q	$p \Rightarrow q$	$q \Rightarrow p$	$(p \Rightarrow q) \vee (q \Rightarrow p)$
T	T	T	T	T
T	F	F	T	T
F	T	T	F	T
F	F	T	T	T

So, a proposition or its converse must always be true.

2.3.2. Contradiction:

A compound proposition that is **false in all possible** interpretations of its components is called a **Contradiction**. It is the negation of a tautology.

Example 2

Show that the compound proposition $(p \wedge \neg p)$ is a contradiction.

p	$\neg p$	$(p \wedge \neg p)$
T	F	F
F	T	F

Thus, a proposition and its negation may not be true at the same time.

2.3.3. Contingency:

A compound proposition that is neither a tautology nor a contradiction is called a **contingency** proposition.

2.4. Further Examples

Example 1

Using the statements p and q for ‘Joseph plays football’ and ‘Joseph is fit’, respectively, write the following statement in symbolic form.

- i. Joseph does not play football. $\neg p$
- ii. Joseph plays football and he is fit. $p \wedge q$
- iii. Joseph plays football or he is fit. $p \vee q$
- iv. If Joseph plays football, then he is fit. $p \Rightarrow q$
- v. Joseph is fit only if he plays football. $p \Leftrightarrow q$.

In order to consider more complex compound propositions, we need to mention rules involving **precedence rules** for logical connectives. Brackets around expressions need to be considered first. The connective $\neg$ has the highest precedence. Therefore, $\neg p \vee q$ is to be understood as $(\neg p) \vee q$ and not $\neg(p \vee q)$. In the case of binary connectives, the highest precedence is given to $\wedge$, followed by $\vee$, $\Rightarrow$ and $\Leftrightarrow$ respectively. Thus, $p \Rightarrow q \vee r$ is to be understood as, $p \Rightarrow (q \vee r)$, and $p \Leftrightarrow p \Rightarrow q$ must be understood as $p \Leftrightarrow (p \Rightarrow q)$, since the connective $\Leftrightarrow$ has the lowest precedence.

If the precedence rules are not sufficient to remove all ambiguities, then brackets must be used.

Example 2

Construct the truth tables for the following compound propositions:

- i. $p \wedge (p \vee q)$
- ii. $(p \Rightarrow q) \wedge (q \Rightarrow p)$
- iii. $[(p \wedge q) \vee r] \Leftrightarrow r$
- iv. $(\neg p \vee q) \Leftrightarrow (p \Rightarrow q)$

i.

p	q	$(p \vee q)$	$p \wedge (p \vee q)$
T	T	T	T
T	F	T	T
F	T	T	F
F	F	F	F

ii.

p	q	$(p \Rightarrow q)$	$(q \Rightarrow p)$	$(p \Rightarrow q) \wedge (q \Rightarrow p)$
T	T	T	T	T
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

iii.

p	q	r	$p \wedge q$	$[(p \wedge q) \vee r]$	$[(p \wedge q) \vee r] \Leftrightarrow r$
T	T	T	T	T	T
T	T	F	T	T	F
T	F	T	F	T	T
T	F	F	F	F	T
F	T	T	F	T	T
F	T	F	F	F	T
F	F	T	F	T	T
F	F	F	F	F	T

Note that this example contains three propositional variables p , q , and r and since each variable can be assigned two values, T and F , there are $2^3 = 8$ assignments in total. In general, a truth table with n propositional variables has 2^n assignments.

iv.

p	q	$\neg p$	$(\neg p \vee q)$	$(p \Rightarrow q)$	$(\neg p \vee q) \Leftrightarrow (p \Rightarrow q)$
T	T	F	T	T	T
T	F	F	F	F	T
F	T	T	T	T	T
F	F	T	T	T	T



Note that this last compound proposition always has the truth value T , regardless of the logical values of its components – this, therefore, is a tautology.

2.5. Argument

An argument is represented by a compound proposition in which the *premises* together **logically imply** the *conclusion*. Thus if $p_1, p_2, \dots, p_n$ denote the premises and q denotes the conclusion, then an argument can be written as

$$(p_1 \wedge p_2 \wedge p_3 \wedge \dots \wedge p_n) \Rightarrow q$$

An alternative notation for this argument is

$$p_1, p_2, p_3, \dots, p_n \models q$$

- If the hypothesis is **true** and the conclusion is also **true**, then the argument is **true**.
- If the argument is *true* it is said to be a *valid* (or a sound) argument.
- If an argument is *false*, then it is said to be a *fallacy*.

Example

- i. Determine the validity of the argument $p \Rightarrow q, q \Rightarrow r \models p \Rightarrow r$ using a truth table. The argument can be written as $[(p \Rightarrow q) \wedge (q \Rightarrow r)] \Rightarrow (p \Rightarrow r)$, i.e. if p implies q and q implies r , then p implies r . The truth table is:

p	q	r	$p \Rightarrow q$	$q \Rightarrow r$	Premise $[(p \Rightarrow q) \wedge (q \Rightarrow r)]$	Valid $\Rightarrow$	Conclusion $p \Rightarrow r$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	T	F
T	F	T	F	T	F	T	T
T	F	F	F	T	F	T	F
F	T	T	T	T	T	T	T
F	T	F	T	F	F	T	T
F	F	T	T	T	T	T	T
F	F	F	T	T	T	T	T

Since all possible assignments lead to a T in the valid column, the argument is sound (or valid); i.e. a tautology.

- ii. Check the validity of the argument $p \Rightarrow q, \neg p \models \neg q$ using a truth table.

p	q	$(p \Rightarrow q)$	$\neg p$	Premise $(p \Rightarrow q) \wedge \neg p$	Valid $\Rightarrow$	Conclusion $\neg q$
T	T	T	F	F	T	F
T	F	F	F	F	T	T
F	T	T	T	T	F^*	F
F	F	T	T	T	T	T

The argument is a fallacy because, there is an F entry in the valid column (*), when $(p \Rightarrow q)$ and $\neg p$ are both true, but the conclusion $\neg q$ is false.

We emphasize that the validity of an argument does not depend upon the truth values nor the content of the statements appearing in the argument, but upon the particular form of the argument. This is illustrated in the following example.

Example 15: Consider the following argument:

$S1$: If a man is a bachelor, he is unhappy.

$S2$: If a man is unhappy, he dies young.

.....

S : Bachelors die young.

Here the statement S below the line denotes the *conclusion* of the argument, and the statements $S1$ and $S2$ above the line denote the *premises*.

We claim that the argument $S1, S2 \models S$ is valid. For the argument of the form

$$p \Rightarrow q, q \Rightarrow r \models p \Rightarrow r$$

where p is "He is a bachelor", q is "He is unhappy" and r is "He dies young"; by example 14 this argument is valid.

2.6. Logical Equivalence

Two propositions $P(p, q, \dots)$ and $Q(p, q, \dots)$ are said to be **logically equivalent**, or simply **equivalent or equal**, denoted by $P(p, q, \dots) \equiv Q(p, q, \dots)$ if they have identical truth tables.

Or If $P \models Q$ and $Q \models P$ then $P \equiv Q$

Example

Consider the truth table of $\neg(p \wedge q)$ and $\neg p \vee \neg q$ appearing below.

P	q	$p \wedge q$	$\neg(p \wedge q)$
T	T	T	F
T	F	F	T
F	T	F	T
F	F	F	T

p	q	$\neg p$	$\neg q$	$\neg p \vee \neg q$
T	T	F	F	F
T	F	F	T	T
F	T	T	F	T
F	F	T	T	T

Observe that both truth tables are the same, that is, both propositions are false in the first case and true in the three cases. Accordingly, we can write $\neg(p \wedge q) \equiv \neg p \vee \neg q$.

Other examples will be discussed in the lecture

All the truth tables are based upon the following truth table. I recommend that you memorize this table.

P	q	$\neg p$	$p \wedge q$	$p \vee q$	$p \Rightarrow q$	$p \Leftrightarrow q$
T	T	F	T	T	T	T
T	F	F	F	T	F	F
F	T	T	F	T	T	F
F	F	T	F	F	T	T

2.7. Algebraic Laws with Propositional Logic

<i>LAW</i>	<i>NAME</i>
$P \vee \neg P \equiv T$ $P \wedge \neg P \equiv F$	Complement law Contradiction law
$P \wedge T \equiv P$ $P \vee F \equiv P$	Identity laws
$P \vee T \equiv T$ $P \wedge F \equiv F$	Domination laws
$P \vee P \equiv P$ $P \wedge P \equiv P$	Idempotent laws
$\neg(\neg P) \equiv P$	Double - negation law
$P \vee Q \equiv Q \vee P$ $P \wedge Q \equiv Q \wedge P$	Commutative laws
$P \vee (P \wedge Q) \equiv P$ $P \wedge (P \vee Q) \equiv P$	Absorption laws
$(P \vee Q) \vee R \equiv P \vee (Q \vee R)$ $(P \wedge Q) \wedge R \equiv P \wedge (Q \wedge R)$	Associative laws
$P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$ $P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$	Distributive laws
$\neg(P \vee Q) \equiv \neg P \wedge \neg Q$ $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$	De Morgan's laws

Each of these laws can be proved using truth tables.

We also display some useful equivalences for compound propositions involving conditional statements and biconditional statements.

Logical Equivalences Involving Conditional Statements:

$$\begin{aligned}
 p \Rightarrow q &\equiv \neg p \vee q \\
 p \Rightarrow q &\equiv \neg q \Rightarrow \neg p \\
 p \vee q &\equiv \neg p \Rightarrow q \\
 p \wedge q &\equiv \neg(p \Rightarrow \neg q) \\
 \neg(p \Rightarrow q) &\equiv p \wedge \neg q \\
 (p \Rightarrow q) \wedge (p \Rightarrow r) &\equiv p \Rightarrow (q \wedge r) \\
 (p \Rightarrow r) \wedge (q \Rightarrow r) &\equiv (p \vee q) \Rightarrow r \\
 (p \Rightarrow q) \vee (p \Rightarrow r) &\equiv p \Rightarrow (q \vee r) \\
 (p \Rightarrow r) \vee (q \Rightarrow r) &\equiv (p \wedge q) \Rightarrow r
 \end{aligned}$$

Logical Equivalences Involving Biconditional Statements:

$$\begin{aligned}
 p \Leftrightarrow q &\equiv (p \Rightarrow q) \wedge (q \Rightarrow p) \\
 p \Leftrightarrow q &\equiv (p \wedge q) \vee (\neg p \wedge \neg q) \\
 p \Leftrightarrow q &\equiv \neg p \Leftrightarrow \neg q \\
 \neg(p \Leftrightarrow q) &\equiv p \Leftrightarrow \neg q
 \end{aligned}$$

Example

Use the laws of propositional logic to prove that:

- i. $P \wedge (\neg P \vee Q) \equiv P \wedge Q$.
 $P \wedge (\neg P \vee Q) \equiv (P \wedge \neg P) \vee (P \wedge Q)$ *Distributive*
 $\equiv F \vee (P \wedge Q)$ *Excluded/Contradiction*
 $\equiv P \wedge Q$ *Identity.*
- ii. $P \vee (P \wedge Q) \equiv P$.
 $P \vee (P \wedge Q) \equiv (P \wedge T) \vee (P \wedge Q)$ *Identity*
 $\equiv P \wedge (T \vee Q)$ *Distributive*
 $\equiv P \wedge T$ *Domination*
 $\equiv P$ *Identity*
- iii. $\neg(P \Rightarrow Q)$ and $P \wedge \neg Q$
 $\neg(P \Rightarrow Q) \equiv \neg(\neg P \vee Q)$ *Definition of $P \Rightarrow Q$*
 $\equiv \neg(\neg P) \wedge \neg Q$ *De Morgan law*
 $\equiv P \wedge \neg Q$ *Double negation law*

Also, to check that $P \Rightarrow Q \equiv \neg P \vee Q$, draw up a truth table for both $P \Rightarrow Q$ and $\neg P \vee Q$ and show that they have the same truth values.

iv. Show that $(P \wedge Q) \Rightarrow (P \vee Q)$ is a tautology.

Solution: To show that this statement is a tautology, we will use logical equivalences to demonstrate that it is logically equivalent to **T**.

(Note: This could also be done using a truth table.)

$$\begin{aligned}(P \wedge Q) \Rightarrow (P \vee Q) &\equiv \neg(P \wedge Q) \vee (P \vee Q) && \text{Definition of } P \Rightarrow Q \\ &\equiv (\neg P \vee \neg Q) \vee (P \vee Q) && \text{De Morgan law} \\ &\equiv (\neg P \vee P) \vee (\neg Q \vee Q) && \text{Associative and commutative laws} \\ &\equiv \mathbf{T} \vee \mathbf{T} && \text{Complement law} \\ &\equiv \mathbf{T} && \text{Domination law}\end{aligned}$$

Other examples will be discussed in the lecture

2.8 Relation between Sets and Logic:

If we look to section 1.10 (chapter 1) which describes the Laws of Set Algebra and section 2.7 (chapter 2) which describes Algebraic Laws with propositional Logics we could find that these laws are remarkably similar.

In fact, simply making the following mechanical changes:

- From A, B, C to P, Q, R
- From $\cap$ to $\wedge$
- From $\cup$ to $\vee$
- From c or $'$ to $\neg$
- From $=$ to $\equiv$
- From U to T
- From φ to F

Translate any related expression from Set theory into logical expression (and vice versa).

Example 18

Convert these set theory expressions / equalities to logical notation, using logical variables P, Q , and R .

1. $(A \cup B) \cap (C^c \cup U)$
2. $(A \cap B^c) \cup \varphi = (A^c \cup B^c)$

Solution:

1. $(P \vee Q) \wedge (\neg R \vee T)$
2. $(P \wedge \neg Q) \vee F \equiv (\neg P \vee \neg Q)$

Example 19

Convert these logical expressions / equivalences to set theory notation, using A, B , and C from the universal set U :

1. $\neg(P \wedge Q) \vee (\neg Q \wedge R)$
2. $(P \vee \neg Q) \wedge T \equiv \neg(\neg P \wedge Q)$

Solution:

1. $(A \cap B)^c \cup (B^c \cap C)$
2. $(A \cup B^c) \cap U = (A^c \cap B)^c$

2.9. Construction of Proofs

This is to establish a conclusion from a set of premises. Suppose we wish to obtain some conclusion, Q , when certain premises, $P_1, P_2, \dots, P_n$ are true. We have seen that; this can be done by showing that the implication:

$$P_1 \wedge P_2 \wedge \dots \wedge P_n \mid = Q$$

is a tautology.

In addition, the use of truth tables (which are not manageable for large expressions) and mathematical induction (we will see this later in the module), there are several standard methods for proofs, including the following

2.9.1 Proof by exhaustion –

Show the conclusion is true for all possible cases.

Example

Prove that 2 is the only even prime number less than 10.

Solution:

2 is prime.

4 is not prime $4 \div 2 = 2$.

6 is not prime $6 \div 2 = 3$.

8 is not prime $8 \div 2 = 4$.

$\Rightarrow$ 2 is the only even prime number less than 10.

Other examples will be discussed in the lecture

2.9.2 Proof by deduction (inference) or Direct Proof

Example

Prove $A \cap B \subseteq A \cup B$.

Solution:

If $x \in (A \cap B)$ then $x \in A$ (and $x \in B$).

If $x \in A$ then $x \in (A \cup B)$.

$\Rightarrow A \cap B \subseteq A \cup B$.

Other examples will be discussed in the lecture

2.9.3 Indirect Proof (e.g. by contradiction)

We want to show that p is true. We assume it is not and therefore $\neg p$ is true and then derive a contradiction.

Example

Prove that 2 is the only even prime.

Proof:

2 is prime (factors of 2 are 1 and 2).

Assume p is an even prime > 2 .

Since p is even, then 2 divides p .

Therefore p is not prime.

The assumption produces a contradiction and must be false so there is no even prime greater than 2. Hence, 2 is the only even prime.

Example: Prove that $\sqrt{2}$ is irrational.

Proof: We have seen previously that every integer is either even or odd. That is, for every $n \in \mathbb{Z}$ there exists $k \in \mathbb{Z}$, such that either $n = 2k$ or $n = 2k+1$. Now, if $n = 2k$ then $n^2 = (2k)^2 = 4k^2 = 2 \cdot (2k^2)$, which means that if n is even then n^2 is also even. On the other hand, if $n = 2k+1$ then $n^2 = (2k+1)^2 = 4k^2 + 4k + 1 = 2 \cdot (2k^2 + 2k) + 1$, so, if n is odd then n^2 is also odd.

We now proceed with a proof by **contradiction**. Assume that $\sqrt{2}$ is rational. that is, $\sqrt{2} \in \mathbb{Q}$. (This is the assumption that should lead to a contradiction.) By definition, this means that there exist two numbers $p, q \in \mathbb{Z}$, with $q \neq 0$, such that

$$\frac{p}{q} = \sqrt{2} \text{ and thus, } \left(\frac{p}{q}\right)^2 = 2 \Rightarrow \frac{p^2}{q^2} = 2 \Rightarrow p^2 = 2q^2$$

We can assume that p and q have no common divisor, since all common divisors can be divided out to begin with. We have $p^2 = 2q^2$.

This shows that p^2 is even, and consequently p must be even; that is, $p = 2k$ for some $k \in \mathbb{Z}$. Then $p^2 = 4k^2 = 2q^2$, so $2k^2 = q^2$.

This shows that q^2 is even, and consequently that q is even. Thus, both p and q are even, contradicting the fact that p and q have no common divisor. We have reached a contradiction, which completes the proof.

Other examples will be discussed in the lecture

The most common types of proofs are ones where we wish to establish the truth of a proposition of the form $(P \Rightarrow Q)$. For propositional logic there are several different proof methods. For example, using the rules of natural deduction or the formulae of semantic tableau. The latter is a method based on showing that the expressions are tautologies (or contradictions) by examining the structure of the expression using various truth formulae called axioms.

2.9.4 Proof by Induction

To prove that $P(n)$ is true for any nonnegative integer $n \geq n_0$. The steps of mathematical induction are as follows:

- i. Show that $P(n_0)$ is true (Basic of induction)
- ii. Assume $P(n)$ is true (Induction hypothesis)
- iii. Show that $P(n+1)$ is true (Induction step)

Example:

Use the technique of mathematical induction to show that

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}, \quad n \geq 1$$

Proof:

Let $P(n)$: $1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$, then

a)(Basic induction) $P(1)$: $1 = \frac{1(1+1)}{2}$. That is, $P(1)$ is true.

b)(Induction hypothesis) Assume $P(n)$ is true. That is,

$$P(n): 1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

c)(Induction step) We must show that $P(n+1)$: $1 + 2 + 3 + \cdots + n = \frac{(n+1)(n+2)}{2}$,

$$1 + 2 + 3 + \cdots + n + (n + 1) = \frac{n(n+1)}{2} + (n + 1) = \frac{(n+1)(n+2)}{2},$$

Other examples will be discussed in the lecture

EXERCISE 3 – PROPOSITIONAL LOGIC (1)

1. Which of the following are propositions?

- a) $5 > 6$
- b) $(3 + 6 - 2) = 10$
- c) Will you go out?
- d) Seven is a prime number.
- e) Pigs can fly.
- f) The Greeks had a word for it.

2. Given the propositions:

i.

p : It is cold.

q : It is raining.

Write down a simple sentence for each of the following:

(a) $\neg p$

(b) $p \wedge q$

(c) $p \vee q$

(d) $q \vee \neg p$

(e) $\neg p \wedge \neg q$.

ii.

p : It is snowing.

q : The street is slippery.

r : My car has snow tyres.

Formulate each of the following symbolic expressions in words;

(a) $p \wedge q$

(b) $q \wedge r$

(c) $\neg p \vee q$

(d) $\neg q \vee r$

(e) $(p \vee q) \wedge \neg r$

(f) $\neg(\neg q)$

(g) $\neg(p \vee q)$

(h) $\neg(p \wedge q)$.

3. Let p , q , and r be propositions. If p is false, q is true and r is false, determine the truth values of the following compound propositions.

(a) $p \vee q$

(b) $\neg p \vee \neg q$

(c) $(p \wedge q) \vee \neg r$

(d) $(p \vee q) \wedge (\neg p \vee \neg q)$

(e) $[(\neg p \wedge \neg q) \vee r] \wedge [(\neg q \vee \neg p) \wedge \neg r]$

4. Construct truth tables for the following symbolic expressions.

(a) $\neg p \wedge q$

(b) $\neg (p \wedge q)$

(c) $\neg(p \vee \neg q)$

(d) $\neg (p \vee r) \wedge \neg (\neg p \wedge q)$

(e) $(p \wedge q) \vee (\neg p \wedge q) \vee (p \wedge \neg q) \vee (\neg p \wedge \neg q)$. (f) $((p \Rightarrow q) \wedge (q \Rightarrow r)) \Rightarrow (p \Rightarrow r)$

(g) $\neg (p \wedge q) \Leftrightarrow (\neg p \vee \neg q)$.

5. Identify all atomic propositions in the following sentences, symbolize them using letters p , q , r , etc. and convert them into symbolic propositional logic.

(a) Joseph is in the gym, and Joseph likes sport.

(b) Maria has a good car but she is a not good driver.

(c) $x \in A$ or $x \in B$.

(d) $x \in A$ or $x \notin B$.

(e) The team is happy only when they win the race.

(f) I'll work tonight, or I'll stay in and watch television.

6. Using the propositions

p : Logic is easy.

q : There is a logic question.

r : I pass the exam.

Symbolise the following expressions and construct truth tables for the propositions.

(a) If logic is easy then I pass the exam.

(b) If I pass the exam then logic is easy.

(c) I fail the exam if logic is difficult.

(d) Absence of a logic question means I pass the exam.

(e) If logic is difficult but there is no logic question, then I pass the exam.

(f) Logic is difficult. I pass the exam if there is no logic question.

(g) Either logic is easy and I pass the exam, or logic is difficult and I fail the exam.

(h) If logic is easy, then if there is a logic question I pass the exam.

(i) If logic is difficult then I pass the exam if there is no logic question.

(j) If I fail the exam, then there is a logic question and logic is difficult.

7. Establish whether each of the following logical expressions is a tautology, a contradiction or a contingency using truth tables.

$$(a) (p \wedge q) \Rightarrow (p \vee q)$$

$$(b) \neg(p \wedge q) \vee (\neg p \vee \neg q)$$

$$(c) (p \Rightarrow q) \Rightarrow (\neg p \vee q)$$

$$(d) [\neg p \wedge (p \Leftrightarrow q)] \Rightarrow \neg q$$

$$(e) [(p \Rightarrow q) \wedge p] \Rightarrow q$$

$$(f) (p \Leftrightarrow q) \Leftrightarrow (p \Leftrightarrow \neg q).$$

$$(g) p \Rightarrow q \wedge (p \Rightarrow \neg q)$$

$$(h) [(p \Rightarrow q) \wedge r] \Leftrightarrow [p \Rightarrow (q \wedge r)].$$

8. Use truth tables to demonstrate the logical equivalence of the following.

$$(a) p \vee (p \wedge q) \equiv p$$

$$(b) \neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$(c) (p \wedge q) \Rightarrow r \equiv (p \Rightarrow r) \vee (q \Rightarrow r)$$

$$(d) p \Rightarrow (q \wedge r) \equiv (p \Rightarrow q) \wedge (p \Rightarrow r)$$

9. Write the converse, inverse and contrapositive for the following:

(a) If it is cold, then it will snow.

(b) If you don't practice, you will never learn maths.

10. Determine the validity of the arguments.

$$(a) p \Leftrightarrow q, q \models p$$

$$(b) p \Rightarrow q, \neg q \models \neg p.$$

11. Use the laws of propositional logic to simplify the following expressions.

$$(a) \neg(\neg p \wedge \neg q)$$

$$(b) \neg(p \Rightarrow \neg q)$$

$$(c) \neg[\neg p \vee (p \Rightarrow \neg q)]$$

$$(d) (p \Rightarrow q) \wedge (q \Rightarrow \neg q)$$

$$(e) p \wedge (p \vee \neg q)$$

$$(f) p \vee (p \wedge q).$$

12. The following are some of the rules of inference. Show their validity by using truth tables.

$$(a) P, Q \models P \wedge Q \quad (\text{Combination law}).$$

$$(b) P \wedge Q \models Q \quad (\text{Simplification law}).$$

$$(c) P, P \Rightarrow Q \models Q \quad (\text{Detachment law, or Modus ponens - method of affirming}).$$

$$(d) \neg Q, P \Rightarrow Q \models \neg P \quad (\text{Contrapositive law or Modus tollens - method of denying}).$$

$$(e) P \Rightarrow Q, Q \Rightarrow P \models P \Leftrightarrow Q \quad (\text{Equivalence introduction}).$$

13. Write a proof that the sum of two even integers is even.

14. Write a proof that if n is even, then n^2 is divisible by 4.

15. Convert $(\neg(P \vee F) \wedge (\neg Q \wedge T)) \vee P$ to set notation, using sets A and B .

16. Convert $(A^c \cup \varphi) \cap (B \cup C)^c = A^c \cap B^c \cap C^c$ to logical notation, using logical variables P, Q , and R .

17. Verify the logical equivalence $P \vee (\neg P \wedge Q) \equiv P \vee Q$

DISCRETE MATHEMATICS

CHAPTER 2 – LOGIC (2)

2.10. Predicates

Predicate logic is an extension of propositional logic with more expressive power. It is needed because the type of argument captured by propositional logic is limited. Many statements in mathematics are not simply propositions. In many instances there are statements whose truth depends on the value of a variable that occurs in the statement.

Example 1

- i. $(10 > 0)$ – a proposition, – true.
- ii. $(x > 0)$ – not a proposition, – the truth value depends upon x .
- iii. ‘ x is greater than y ’, – not a proposition,
– the truth value of this statement depends on the values of x and y .
For some values of x and y (e.g. $x = 3, y = 1$ say) the logical value of the expression is true but for some other values it (e.g. $x = 1, y = 5$ say) it is false.

Definition:

A predicate or an open statement is a statement containing a variable (or an argument) such that when the variable is substituted by an acceptable specific **object** the statement becomes a proposition.

A set of all acceptable objects that can be permitted to replace the variable is called the **domain** (or the universe of discourse) of the predicate or the replacement set of the variable(s).

Alternatively, we can define a predicate as being **a function**, taking arguments (or parameters) and returning a value true or false.
OR

A predicate $P(x)$ is a statement that incorporates a variable x , such that whenever x is replaced by a value, the resulting proposition is unambiguously true or false. If the predicate uses several variables $x_1, x_2, x_3, \dots$ then we will extend the notation to $P(x_1, x_2, \dots)$ accordingly.

Note that in the literature several other words are used for the word *object* including ‘subject’, ‘term’ and ‘individual’.

Example 2

Consider the sentence: ‘Manchester and Liverpool are cities’

The descriptive phrase ‘**are cities**’ is a **predicate**. The entities connected in this way, Manchester and Liverpool are the variables.

Example 3

The following statements are all predicates:

- i. She arrived yesterday.
'She' is not identified, and it is not possible to assign a logical value to this expression.
- ii. A is a number greater than 100.
The subject 'A' is not identified and therefore it is not possible to find the logical value of this expression.
- iii. x is a student.
' x ' is not determined and so this expression also has no logical value.

Note that, the *domain* is somewhat arbitrary and so should always be clearly specified. This is similar to the concept of the universal set. Also, it is usual to think only of the elements of the domain for which the predicate is true but remember that any domain element can replace the variable, possibly resulting in a predicate which is false.

The predicates are generally denoted by capital letters and the associated variables or the arguments are denoted by lower case letters listed in parentheses.

Example 4

If $P(n)$ is the predicate " n is even" write the proposition that results when the variable n is replaced by each of the values 2, 17, and 240. Which of the resulting statements are true?

Solution:

We will write $P(2)$ to denote the proposition resulting from replacing n with 2 in the predicate $P(n)$. In this case, $P(2)$ is the true proposition "2 is even". Similarly, $P(17)$ denotes replacing n by 17, giving us the false statement "17 is even". Finally, $P(240)$ is the true statement "240 is even".

Example 5

- i. $P(x)$, with $x \in N$
Here P is the predicate, x is the variable and N is the domain.
- ii. $P(x, y)$, with $x, y \in R$
This is a predicate with two variables x and y and R is the domain.



In the notation used the predicate P is a function which takes arguments (variables x and/or y) and returns a (Boolean) value true or false.

Other examples will be discussed in the lecture

2.11 Generating propositions:

There are two methods of generating propositions from predicates

- INSTANTIATION;

and

- QUANTIFICATION.

2.11.1 Instantiating Predicates

This is to substitute values for the variable of the predicate and thus generate propositions and determine the logical value of the resulting proposition.

Example 6

Consider predicate $P(x)$ with $x \in \mathbb{N}$.

That is, the predicate $P(x)$ for integer $x > 0$ is positive and has a logical value true.

Therefore,

$P(3)$ becomes a proposition with value true ($3 > 0$),

$P(-2.5)$ becomes a proposition with value false ($-2.5 < 0$).

The set A is called the domain of $P(x)$, and the set T_p of all elements of A for which $P(a)$ is true is called the *truth set* of $P(x)$. In other words,

$$T_p = \{x: x \in A, P(x) \text{ is true}\} \text{ or } T_p = \{x: P(x)\}$$

Frequently, when A is some set of numbers, the condition $P(x)$ has the form of an equation or inequality involving the variable x .

Example 7

Find the truth set T_p of each propositional function $P(x)$ defined on the set $P = \{1, 2, 3, \dots\}$.

a) Let $P(x)$ be " $x+2 > 7$ ". Then $T_p = \{x: x \in P, x + 2 > 7\} = \{6, 7, 8, \dots\}$, and it consists of all integers greater than 5.

b) Let $P(x)$ be " $x+5 < 3$ ". Then $T_p = \{x: x \in P, x + 5 < 3\} = \varnothing$, and it is the empty set.

In other words, $P(x)$ is not true for any positive integer in P .

c) Let $P(x)$ be " $x+5 > 1$ ". Then $T_p = \{x: x \in P, x + 5 > 1\} = P$

Thus $P(x)$ is true for every element in P .

2.11.2 Quantifying Predicates

Quantifiers indicate how frequently a certain statement is true. When the variable say x of a predicate $P(x)$ is replaced by some element belonging to a set say A , the resulting proposition is either true or false. Sometimes the logical value of a predicate is true (or false) for all x or some x belonging to a set A .

The phrases '*for all x* ', '*for each x* ' and '*for any x* ' are referred to as the **universal quantifier** and the phrases '*for some x* ', '***there exists an element x*** ' and '***there are x and y*** ' are referred to as the **existential quantifier**. The set A is called the *scope* of the quantifier. The variable x is said to be *bound* by the quantifier.

In symbolic form when using the *universal quantifier* we write $\forall x \in A$, which is read as 'for all x belonging to A '. The symbol $\forall$ is read 'for all'.

Similarly, when using the *existential quantifier* we write $\exists x \in A$, which is read as 'for some x belonging to A '. The symbol $\exists$ is read 'for some' or 'there exists'. For the existential quantifier other expressions such as 'there exists an x such that' and 'there exists at least one x such that' are also used.

Example 8

Express the following in symbolic predicate logic.

- i. 'Everyone gets a break once in a while'.

Define P to mean 'gets a break once in while'.

Thus, the predicate $P(x)$ means that person x gets a break once in a while.

The word '*everyone*' indicates that this is true *for all x* .

Therefore, in symbolic form the expression is $\forall x, P(x)$.

- ii. 'Some students have a car'.

Define P to mean 'have a car'.

Thus $P(x)$ means that student x has a car.

The word '*some*' indicates that this is true *for some x* .

Therefore, in symbolic form the expression is $\exists x, P(x)$.

Example 9

Let $D = \{3, 4, 5, 10, 20, 25\}$. Express each sentence using the symbols $\forall$ and $\exists$ and decide whether or not it is true.

- For every n that is a member of D , $n < 20$.
- For all n in the set D , $n < 5$ or n is a multiple of 5.
- There is (at least one) k in the set D with the property that k^2 is also in the set D .
- There exists m a member of the set D such that $m \geq 3$.

Solution:

- $\forall n \in D, n < 20$. This statement is false, since 20 and 25 are counterexamples.
- $\forall n \in D, n < 5$ or n is a multiple of 5. This statement is true.
- $\exists k \in D$ such that $k^2 \in D$. This statement is true.
- $\exists m \in D$ such that $m \geq 3$. or more simply, just $\exists m \in D, m \geq 3$. this statement is true.

Example 10

Given that P is the set of **natural numbers** consider when the following propositions are true or false:

- a) The proposition $\forall x \in P, x+4 > 3$.
It is true since $\{x: x+4 > 3\} = \{1, 2, 3, \dots\} = P$.
- b) The proposition $\forall x \in P, x+2 > 8$.
It is false since $\{x: x+2 > 8\} = \{7, 8, \dots\} \neq P$.
- c) The proposition $\exists x \in P, x+4 < 7$.
It is true since $\{x: x+4 < 7\} = \{1, 2\} \neq \varnothing$.
- d) The proposition $\exists x \in P, x+6 < 4$.
It is false since $\{x: x+6 < 4\} = \varnothing$.

2.11.3 Quantifying Predicates with two variables:

If $G(x, y)$ is the predicate $(x > y)$ then quantifying over both variables produces a proposition

$$\forall x: N. (\exists y: N. G(x, y))$$

'Every natural number is greater than some natural number'.

This is not the same as

$$\exists y: N. (\forall x: N. G(x, y))$$

'There is a natural number that all-natural numbers are greater than'

Example 11

Let $P(x, y)$ be the predicate $x \cdot y = 36$.

- a) Identify which of the following statements are true; $P(3, 4)$, $P(-9, -4)$, $P(12, -1)$.
- b) Assuming the domain is the set of integers, find other values for x and y that make the statement $P(x, y)$ true.

Solution:

- a) The statements are $3 \cdot 4 = 36$, $(-9) \cdot (-4) = 36$, and $12 \cdot (-1) = -12$. Only the second statement is true, the first and third are false.
- b) Here are some additional true statements: $P(36, 1)$, $P(6, 6)$ and $P(-2, -18)$.
There are several more that are true.

2.11.4 Alternative Notation

Note that some books use the notation $\forall x. P(x)$ or simply $\forall x. P$ (instead of $\forall x \in P(x)$) and use $\exists x. P(x)$, or $\exists x. P$ (instead of $\exists x \in P(x)$). To include the domain, the notations

$\forall x: N. P(x)$ and $\exists x: N. P(x)$ are also used.

For two variables $\forall x \forall y. P(x, y)$

Example 12

Consider the predicate $P(x) : x > 0$, then

- i. $\forall x: N . P(x)$ reads as 'All natural numbers are positive'.
- ii. $\exists x: R . \neg P(x)$ reads as 'there exists a real number which is not positive'.
- iii. $\neg (\exists x: N . \neg P(x))$ reads as 'There is no natural number which is not positive'.
- iv. $\neg (\forall x: R . P(x))$ reads as 'Not all real numbers are positive'.

2.12. Negations of quantifiers

The following rules apply for the negation of quantified statements:

- Replace $\neg \forall x . P$ by $\exists x . \neg P$ (i.e. replace $\neg \forall$ by $\exists$ and P by $\neg P$).
- Replace $\neg \exists x . P$ by $\forall x . \neg P$ (i.e. replace $\neg \exists$ by $\forall$ and P by $\neg P$).

In other words: if $P(x)$ is a propositional function with variable x , then

- a) $\neg [(\forall x) . P(x)]$ is equivalent to $(\exists x) \neg [P(x)]$.
(i.e. It is not true, that for all x , $P(x)$ is true **is equivalent to** there exists an x such that $P(x)$ is false).
- b) $\neg [(\exists x).P(x)]$ is equivalent to $(\forall x) \neg [P(x)]$.
(i.e. It is not true for some x , $P(x)$ is true **is equivalent to** For all x , $P(x)$ is false).

Example 13

- i. $(\forall x: Z) [x < 3]$ becomes $(\exists x: Z) [x \geq 3]$.
i.e. all integer numbers are less than 3.
The negation becomes:
There exists some integer numbers which are ≥ 3 .
- ii. $(\exists x: N/\{1\}) [x^3 \leq x^2]$ becomes $(\forall x: N/\{1\}) [x^3 > x^2]$.
i.e. there exists some natural numbers such that their cubes are less than or equal to their squares.
The negation becomes:
The cubes of all-natural numbers (excluding 1) are greater than their squares.

2.13 Relationship between quantifiers and sets

For a finite set $S = \{e_1, e_2, \dots, e_n\}$, the following two relationships are defined:

(1) $\forall x \in S \cdot P(x) \equiv P(e_1) \wedge P(e_2) \wedge \dots \wedge P(e_n)$.

Is the logical *and* of all the propositions generated by instantiating x with each member of S .

(2) $\exists x \in S \cdot P(x) \equiv P(e_1) \vee P(e_2) \vee \dots \vee P(e_n)$.

Is the logical *or* of all the propositions generated by instantiating x with each member of S .

Thus, in terms of truth sets:

- i) $\neg P(x)$ is the *complement* of $P(x)$.
- ii) $P(x) \wedge Q(x)$ is the *intersection* of $P(x)$ and $Q(x)$.
- iii) $P(x) \vee Q(x)$ is the *union* of $P(x)$ and $Q(x)$.

2.14 Laws of predicate logic

All the propositional laws hold for predicate logic, including truth tables etc., but we need additional laws in predicate logic to deal with quantifiers.

The following equivalences involving quantifiers are a few examples of such laws.

- Negations through Quantifiers.

$$\begin{aligned}\neg \forall x . A &\equiv \exists x . \neg A \\ \neg \exists x . A &\equiv \forall x . \neg A\end{aligned}$$

- Universal quantifiers across conjunction and existential across disjunction.

$$\begin{aligned}\forall x . (A \wedge B) &\equiv \forall x . A \wedge \forall x . B \\ \exists x . (A \vee B) &\equiv \exists x . A \vee \exists x . B\end{aligned}$$

- Interchanging two universal and two existential quantifiers.

$$\begin{aligned}\forall x \forall y . A &\equiv \forall y \forall x . A \\ \exists x \exists y . A &\equiv \exists y \exists x . A\end{aligned}$$

Example 14

- a) Consider the quantified statement “*There exist integers x and y such that $x \cdot y = 36$* ”. Write this using symbolic logic notation, and decide if it is true or not.
- b) Do the same for the quantified statement “*For all integers x and y , it is true that $x \cdot y = 36$* ”.

Solution:

- a) We can write this in two equivalent ways:

- $\exists x \in \mathbb{Z}, \exists y \in \mathbb{Z}, x \cdot y = 36$
- $\exists x, y \in \mathbb{Z}, x \cdot y = 36$.

The latter is a shortcut that means the same as the former. In either case, we can use $P(x, y)$ to indicate the predicate $x \cdot y = 36$, and write $\exists x \in \mathbb{Z}, \exists y \in \mathbb{Z}, P(x, y)$, or $\exists x, y \in \mathbb{Z}, P(x, y)$. The quantified predicate is true since, for example, $P(-2, -18)$ is true.

- b) Similarly, we can write the “for all” quantifiers in two equivalent ways:

- $\forall x \in \mathbb{Z}, \forall y \in \mathbb{Z}, x \cdot y = 36$
- $\forall x, y \in \mathbb{Z}, x \cdot y = 36$.

Since $P(10, 2)$ is the false statement “ $10 \cdot 2 = 36$ ”, we see that the quantified predicate is false.

Example 15

Translate the statement ‘Everyone here speaks English and French’ into predicate logic in two ways. The Domain is supposed to be a group of people.

Let $E(x)$: x speaks English, $F(x)$: x speaks French.

Then in predicate logic this is $\forall x \cdot (E(x) \wedge F(x))$,

or simply $\forall x \cdot (E \wedge F) \equiv \forall x \cdot E \wedge \forall x \cdot F$, using Universal quantifiers across conjunction.

i.e. Everyone here speaks English and French $\equiv$ Everyone here speaks English and Everyone here speaks French.

2.15 The Truth Set of Predicates

The set of all variables for which a predicate is true is called the truth set of the predicate. Thus, the truth set (say set A) of a predicate $P(x)$, is the subset of the domain specified by: $A = \{x / P(x) \text{ is true}\}$.

Example 16

Find the truth set for the predicate:

$$S(x) : (x - 3)(x + 2)(x - 1) = 0, \quad x \in N.$$

Solution:

The truth set is $A = \{1, 3\}$, which is a subset of N . Notice that -2 is not an element of the truth set because it does not belong to the domain N .

Example 17

Let $A = \{1, 2, 3, 4, 5\}$. Determine the truth set of each of the following statements:

- a) $(\exists x \in A)(x + 3 = 10)$
- b) $(\forall x \in A)(x + 3 < 10)$
- c) $(\exists x \in A)(x + 3 < 5)$
- d) $(\forall x \in A)(x + 3 \leq 7)$

Solution:

- a) False. φ (For no number in A is a solution to $x + 3 = 10$).
- b) True. A (for every number in A satisfies $x + 3 < 10$).
- c) True. $\{1\}$.
- d) False. φ .

EXERCISE 4 – PREDICATE LOGIC (2)

1. Identify the following expressions as a proposition or a predicate. In the case of a predicate indicate the object and the predicate for each expression:
 - a. John goes to university.
 - b. She has a cat.
 - c. Most computer users like the Internet.
 - d. x is an even number greater than 10.
2. Express the following statements in predicate logic. The universe of discourse is the set of all persons.
 1. If Mary likes Steve, and Steve likes Jane, then Mary likes Jane.
 2. John is very busy, but Gary is not.
 3. I know you, but you don't know me.
3. Convert each of the following statements into universally quantified propositions in symbolic form, negate them and then express the negations in words.
 - (a) All blue cars are slow.
Let $C(x) = x$ is a car. $B(x) = x$ is blue. $S(x) = x$ is slow.
 - (b) Every computing student does maths.
Let $C(x) = x$ is a computing student. $M(x) = x$ is Student does maths.
 - (c) Each module in the course has twenty credits.
Let $C(x) = x$ is a module on the course. $T(x) = x$ has twenty credits.
4. Determine the logical value of the following expressions if $x = 2$ and $y = 10$.
 - (a) $((x = 3) \text{ and } (y < 9) \text{ or } (y \geq 4))$.
 - (b) $(\text{not } ((x = 3) \text{ and } (y \geq 4)))$.
5. Assume that x, y are real numbers and let $P(x, y)$ denote the predicate $x + y = 0$. Express each of the following expressions in words and evaluate the truth value.
 - (g) $\forall x (\exists y (P(x, y)))$.
 - (h) $\exists x (\forall y (P(x, y)))$.
6. In the following, x is a number, c is a character and s and t are strings. Define the predicate, $S = \text{starts}(x, y)$ for parts (b) and (c). Express the following sentences in predicate logic.
 - (a) x is either less than two or greater than five but less than ten. However, x is not seven.
 - (b) If s starts with c then t starts with c .
 - (c) s starts with the same character as t .

7. Let $Cars$ = set of cars. $R(x)$: x is red. $F(x)$: x is fast. $C(x)$: x is a car.
Express each of the following propositions in words.
Note that in the following, ‘:’ means ‘ $\in$ ’.
- $\neg \forall x : Cars \cdot F(x)$
 - $\forall x : Cars \cdot (R(x) \rightarrow F(x))$
 - $\exists x : Cars \cdot \neg F(x)$
 - $\neg \exists x : Cars \cdot (R(x) \wedge \neg F(x))$
 - $\forall x : Cars \cdot (\neg R(x) \rightarrow F(x))$
 - $\forall x : Cars \cdot (F(x) \rightarrow R(x))$
 - $\neg \exists x : Cars \cdot (F(x) \wedge \neg R(x))$
 - $\neg \exists x : Cars \cdot (\neg R(x) \wedge \neg F(x))$.
8. Evaluate and give an interpretation of each of the following assertions.
- $\forall x : \mathbb{R} \cdot (\exists y : \mathbb{R} \cdot (x \cdot y = 1))$
 - $\forall x : \mathbb{R} \cdot (\exists y : \mathbb{R} \cdot (x + y = 0))$
 - $\forall x : \mathbb{N} \cdot (\exists y : \mathbb{N} \cdot (x \cdot y = 1))$
 - $\forall x : \mathbb{N} \cdot (\exists y : \mathbb{N} \cdot (x + y = 0))$
 - $\forall x : \mathbb{Z} \cdot (\exists y : \mathbb{Z} \cdot (x \cdot y = 1))$
 - $\forall x : \mathbb{Z} \cdot (\exists y : \mathbb{Z} \cdot (x + y = 0))$.
9. If S is the set of students in a class and the following predicates are defined,
 $P(x)$: x is sitting the exam.
 $Q(x)$: x has passed the continuous assessment.
Give an English interpretation of the following:
- $\forall x \in S \cdot P(x) \Rightarrow Q(x)$.
 - $\neg \exists x \in S \cdot (P(x) \wedge \neg Q(x))$.
 - Using the laws of predicate logic show that the expressions in (a) and (b) are equivalent.
 - Give a formal representation of: ‘there is at least one student not sitting the exam who has passed the continuous assessment.’
 - Give in words an interpretation of the formal expression:
 $\neg \forall x \in S \cdot P(x) \Rightarrow Q(x)$.
10. For sets A and B drawn from some Universal set, U , the following predicates are defined:
contained $(A, B) : A \subseteq B$.
equal $(A, B) is : A = B$

Express in English and decide which are always true. Give examples for those which are false.

- $(\text{contained}(A, B) \wedge \text{contained}(B, C)) \Rightarrow \text{contained}(A, C)$
- $(\text{contained}(A, B) \wedge \text{contained}(B, A)) \Leftrightarrow \text{equal}(A, B)$
- $(\text{contained}(A, B) \wedge \neg(\text{equal}(A, B))) \Rightarrow \text{contained}(B, A)$.

11. Let P be the set of people. The following predicates are defined:

$x, y \in P$.

$wife(x, y) : x \text{ is } y\text{'s wife.}$

$male(x) : x \text{ is male.}$

$brother(x, y) : x \text{ and } y \text{ are brothers.}$

$child(x) : x \text{ is a child.}$

$female(x) : x \text{ is female.}$

$parents(x, y, z) : x \text{ and } y \text{ are parents of } z.$

i. Express in English:

(a) $\forall x \in P. \forall y \in P. brother(x, y) \Rightarrow male(x).$

(b) $wife(Susan, John) \wedge \forall x \in P. child(x, Susan) \Rightarrow child(x, John).$

ii. Express in formal notation:

(a) If x is y 's wife then x is female and y is male.

(b) Susan is John's wife, but John has at least one child who is not Susan's.

(c) All of Mike and Nicky's children are boys.

(d) One of a person's parents is male and the other is female.

DISCRETE MATHEMATICS

Chapter 3: Relations and Functions

3.0 Introduction

Relations and its types concepts are one of the important topics of set theory.

Sets, relations and functions all three are interlinked topics. Sets denote the collection of ordered elements whereas relations and functions define the operations performed on sets.

A **relation** defines the connection between the two given sets. Also, there are types of relations stating the connections between the sets. Hence, here we will learn about relations and its types in detail.

3.1 Definition: Relation

A relation in mathematics defines the relationship between two different sets of information. If two sets are considered, the relation between them will be established if there is a connection between the elements of two or more non-empty sets.

Therefore, we can say,

‘A set of ordered pairs is defined as a relation “ R ”.’

So, a binary relation R from a set A to a set B is a subset of the cartesian product between the sets A and B ($A \times B$). (i.e. $R \subset A \times B$). If $(a,b) \in A \times B$ we write aRb .

The set $Dom(R) = \{a \in A \mid (a,b) \in R \text{ for some } b \in B\}$ is called the **domain of R** .

The set $Range(R) = \{b \in B \mid (a,b) \in R \text{ for some } a \in A\}$ is called the **range of R** .

Example: For ordered pairs = $\{(1,2), (-3,4), (5,6), (-7,8), (9,2)\}$

The domain is $= \{-7, -3, 1, 5, 9\}$

and the range is $= \{2, 4, 6, 8\}$

Example: Let $A = \{2, 3, 4\}$ and $B = \{3, 4, 5, 6, 7\}$. Define the relation R by aRb if and only if a divides b . Find, R , $Dom(R)$ and $Range$ of (R) .

Solution:

$R = \{(2,4), (2,6), (3,3), (3,6), (4,4)\}$

$Dom(R) = \{2, 3, 4\}$ and

$Range(R) = \{3, 4, 6\}$

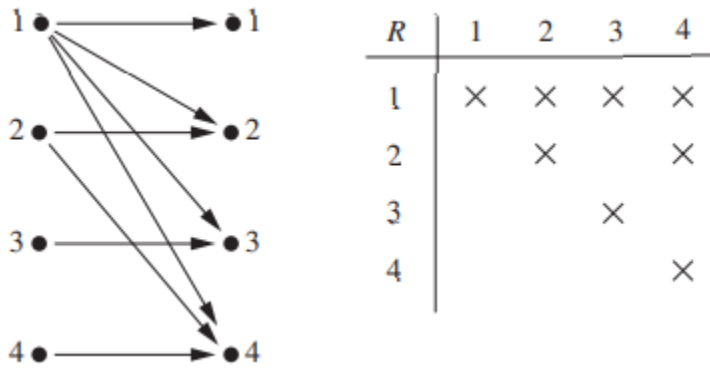
Example: Let A be the set $\{1, 2, 3, 4\}$.

Which ordered pairs are in the relation $R = \{(a, b) \mid a \text{ divides } b\}$?

Solution:

Because (a, b) is in R if and only if a and b are positive integers not exceeding 4 such that a divides b , we see that $R = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 4), (3, 3), (4, 4)\}$.

The pairs in this relation are displayed both graphically and in tabular form in the Figure below.



Example:

Consider these relations on the set of integers:

$$R1 = \{(a, b) \mid a \leq b\},$$

$$R2 = \{(a, b) \mid a > b\},$$

$$R3 = \{(a, b) \mid a = b \text{ or } a = -b\},$$

$$R4 = \{(a, b) \mid a = b\},$$

$$R5 = \{(a, b) \mid a = b + 1\},$$

$$R6 = \{(a, b) \mid a + b \leq 3\}.$$

Which of these relations contain each of the pairs $(1, 1)$, $(1, 2)$, $(2, 1)$, $(1, -1)$, and $(2, 2)$?

Solution:

The pair $(1, 1)$ is in $R1$, $R3$, $R4$, and $R6$;

$(1, 2)$ is in $R1$ and $R6$;

$(2, 1)$ is in $R2$, $R5$, and $R6$;

$(1, -1)$ is in $R2$, $R3$, and $R6$;

and finally, $(2, 2)$ is in $R1$, $R3$, and $R4$.

Example: How many relations are there on a set with n elements?

Solution:

A relation on a set A is a subset of $A \times A$. Because $A \times A$ has n^2 elements when A has n elements, and a set with m elements has 2^m subsets, there are 2^{n^2} subsets of $A \times A$.

Thus, there are 2^{n^2} relations on a set with n elements.

For example, there are $2^{3^2} = 2^9 = 512$ relations on the set $\{a, b, c\}$.

3.1.1 Types of Relations

There are 9 main types of relations which include:

- **Reflexive Relation**

In a reflexive relation on a set A , **every element** map to itself.

i.e. the reflexive relation is given by $(a, a) \in R$ for all $a \in A$ (i.e. $\forall a \in A$)

For example, consider a set $A = \{1, 2, 3, 4\}$.

Now an example of reflexive relation will be

$$R = \{(1, 1), (2, 2), (3, 3), (4, 4)\}.$$

An example for a reflexive relation:

(a) $(\forall x \in \mathbb{R} \text{ (Real numbers)}) (x = x)$. (Equality is a reflexive relation).

(b) $(\forall x \in \mathbb{R}) (x < x)$. (Less than is **not** a reflexive relation).

(c) $(\forall x \in \mathbb{R}) (x \geq x)$. (Greater than or equal is a reflexive relation).

(d) $(\forall a \in \mathbb{N}) (a \mid a)$. (Divides is a reflexive relation).

- **Irreflexive Relation**

Irreflexive is given by $(a, a) \notin R$ for all $a \in A$ (i.e. $\forall a \in A$)

An example for irreflexive (**not** a reflexive) relation:

$(\forall x \in \mathbb{R}) (x \neq x)$. (Not equal is not a reflexive relation).

- **Symmetric Relation**

In a symmetric relation on a set A , **if $a R b$ is true then $b R a$ is also true**. In other words, a relation R on a set A is symmetric only if $(b, a) \in R$ is true when $(a, b) \in R$.

i.e. for a symmetric relation on a set A , $a R b \Rightarrow b R a$, for all $a, b \in A$ (i.e. $\forall a, b \in A$)

Example: Let $A = \{1, 2, 3\}$ and $R = \{(1, 1), (2, 2), (1, 2), (2, 1), (2, 3), (3, 2)\}$

Solution: The relation R is symmetric as for every $(a, b) \in R$, we have $(b, a) \in R$,

i.e., $(1, 2), (2, 1), (2, 3), (3, 2) \in R$

An example of symmetric relations:

$(\forall x, y \in \mathbb{R}) ((x = y) \rightarrow (y = x))$ (Equality is symmetric).

An example for not symmetric relations:

- (a) $(\forall x, y \in \mathbb{R}) ((x \neq y) \rightarrow (y \neq x))$. (Not equal is symmetric).
- (b) $(\forall x, y \in \mathbb{R}) (x < y \rightarrow y < x)$. (Less than is **not** symmetric).
- (c) $(\forall x, y \in \mathbb{R}) (x \geq y \rightarrow y \geq x)$. (Greater than or equal is **not** symmetric).
- (d) $(\forall a, b \in \mathbb{N}) (a \mid b \rightarrow b \mid a)$. (Divides is **not** symmetric).
- (e) Relation vertical (**Lr**) is symmetric since a line a is **Lr** to b , then b is **Lr** to a .
- (f) Relation Parallel ($\parallel$) is symmetric, since if a line a is $\parallel$ to b then b is also $\parallel$ to a .

- **Anti-Symmetric Relation**

A relation R on a set A is said to be anti-symmetric relation,

if $(a, b) \in R$ and $(b, a) \in R \Rightarrow a = b, \forall a, b \in A$

Example: Let $A = \{1, 2, 3\}$ and $R = \{(1, 1), (2, 2)\}$.

Is the relation R antisymmetric?

Solution:

The relation R is antisymmetric as $a = b$ when (a, b) and (b, a) both belong to R .

An example for Anti-Symmetric relations:

- (a) $(\forall x, y \in \mathbb{R}) (((x = y) \wedge (y = x)) \Rightarrow (x = y))$, (Equality is anti-symmetric).
- (b) $(\forall x, y \in \mathbb{R}) ((x < y \wedge y < x) \Rightarrow (x = y))$, (Less than is anti-symmetric).
- (c) $(\forall x, y \in \mathbb{R}) (x \geq y \wedge y \geq x \Rightarrow x = y)$, (Greater than is anti-symmetric).
- (d) $(\forall a, b \in \mathbb{N}) (a \mid b \wedge b \mid a \Rightarrow a = b)$, (Divides is anti-symmetric).

An example for not antisymmetric

$(\forall x, y \in \mathbb{R}) ((x \neq y \wedge y \neq x) \Rightarrow (x = y))$. (Not equal is not anti-symmetric)

Example: Let $A = \{4, 5, 6\}$ and $R = \{(4, 4), (4, 5), (5, 4), (5, 6), (4, 6)\}$.

Is the relation R antisymmetric?

Solution:

The relation R is not antisymmetric as $4 \neq 5$ but $(4, 5)$ and $(5, 4)$ both belong to R .

- **Transitive Relation**

A relation R on a set A is called transitive **if** $(a, b) \in R$, $(b, c) \in R$, **then** $(a, c) \in R$, **for all** $a, b, c \in A$.

i.e. R is a transitive relation on a set A , $a R b$ and $b R c \Rightarrow a R c \forall a, b, c \in A$

Example: Let $A = \{1, 2, 3\}$ and $R = \{(1, 2), (2, 1), (1, 1), (2, 2)\}$.

Is the relation R transitive?

Solution: The relation R is transitive as for every (a, b) and $(b, c) \in R$, we have

$(a, c) \in R$ i.e. $(1, 2)$ and $(2, 1) \in R \Rightarrow (1, 1) \in R$

An example for Transitive relation:

(a) $(\forall x, y, z \in \mathbb{R}) ((x = y) \wedge (y = z)) \rightarrow (x = z)$. (Equality is transitive)

(b) $(\forall x, y, z \in \mathbb{R}) ((x < y \wedge y < z) \rightarrow (x < z))$. (Less than is transitive).

(c) $(\forall x, y, z \in \mathbb{R}) (x \geq y \wedge y \geq z \rightarrow x \geq z)$. (Greater than or equal is transitive).

(d) $(\forall a, b, c \in \mathbb{N}) (a \mid b \wedge b \mid c \rightarrow a \mid c)$. (Divides is transitive).

An example for non-Transitive relation:

$(\forall x, y, z \in \mathbb{R}) ((x \neq y \wedge y \neq z) \rightarrow (x \neq z))$. (Not equal is **not** transitive).

Note:

(a) $\perp$ is not transitive since $a \perp b$, $b \perp c$ then it is not true that $a \perp c$.

(b) Since no line is $\parallel$ to itself, we can have $a \parallel b$, $b \parallel a$ but $a \not\parallel a$.
Thus $\parallel$ is not transitive, **but it will be transitive in the plane.**

- **Equivalence Relation**

A relation R on a set A is called an *equivalence relation*, if it is simultaneously reflexive, symmetric and transitive on A .

Example: The relation “is equal to”, denoted “=”, is an *equivalence relation* on the set of real numbers since for any $x, y, z \in \mathbb{R}$:

1. (Reflexivity) $x = x$,

2. (Symmetry) if $x = y$ then $y = x$,

3. (Transitivity) if $x = y$ and $y = z$ then $x = z$.

All of these are true.

Example: The relation “is less than or equal to”, denoted “ $\leq$ ”, is **NOT** an equivalence relation on the set of real numbers. For any $x, y, z \in R$, “ $\leq$ ” is reflexive and transitive but **NOT** necessarily symmetric.

1. (Reflexivity) Of course $x \leq x$ is true since $x = x$.
2. (Symmetry) If $x \leq y$ then it is **not** necessarily true that $y \leq x$.
For example, $5 \leq 7$, but $7 \geq 5$.
3. (Transitivity) If $x \leq y$ and $y \leq z$ then $x \leq z$ since $x \leq y \leq z$.

- **Void (Empty) Relation**

An empty relation (or void relation) is one in which there is no relation between any elements of a set.

For empty relation, $R = \emptyset \subset A \times A$

For example, if set $A = \{1, 2, 3\}$ then, one of the void relations can be $R = \{x, y\}$ where, $|x - y| = 8$.

- *Void Relation $R = \emptyset$ is symmetric and transitive but not reflexive.*

- **Universal Relation**

A universal (or full relation) is a type of relation in which every element of a set is related to each other.

Since, $A \times A \subseteq A \times A$, so $A \times A$ is a relation on A , called the universal relation.

i.e. For universal relation, $R = A \times A$

For example, consider set $A = \{a, b, c\}$. Now one of the universal relations will be $R = \{x, y\}$ where, $|x - y| \geq 0$.

Universal Relation is reflexive, symmetric and transitive. So this is an equivalence relation.

- **Identity Relation**

In an identity relation, every element of a set is related to itself only.

i.e. For identity relation, $I_A = \{(a, a), a \in A\}$

For example, in a set $A = \{a, b, c\}$, the identity relation will be $I_A = \{a, a\}, \{b, b\}, \{c, c\}$.

Identity relation I_A on set A is reflexive, transitive and symmetric. So identity relation I_A is an Equivalence Relation.

- **Inverse Relation**

If A and B are two non-empty sets and R be a relation from A to B , such that

$R = \{(a, b) : a \in A, b \in B\}$, then the inverse of R , denoted by R^{-1} , is a relation from B to A and is defined by $R^{-1} = \{(b, a) : (a, b) \in R\}$

For example, if set $A = \{(a, b), (c, d)\}$, then inverse relation will be $R^{-1} = \{(b, a), (d, c)\}$.

- **Combining Relations:**

Because relations from A to B are subsets of $A \times B$, two relations from A to B can be combined in any way two sets can be combined.

Example:

Let $A = \{1, 2, 3\}$ and $B = \{1, 2, 3, 4\}$.

The relations $R_1 = \{(1, 1), (2, 2), (3, 3)\}$ and $R_2 = \{(1, 1), (1, 2), (1, 3), (1, 4)\}$ can be combined to obtain:

$$R_1 \cup R_2 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (3, 3)\},$$

$$R_1 \cap R_2 = \{(1, 1)\},$$

$$R_1 - R_2 = \{(2, 2), (3, 3)\},$$

$$R_2 - R_1 = \{(1, 2), (1, 3), (1, 4)\}.$$

Example:

Given the following two relations from $A = \{1, 2, 4\}$ to $B = \{2, 6, 8, 10\}$:

aRb if and only if $a|b$.

aSb if and only if $b - 4 = a$.

List the elements of R , S , $R \cup S$, and $R \cap S$.

Solution:

We have $R = \{(1, 2), (1, 6), (1, 8), (1, 10), (2, 2), (2, 6), (2, 8), (2, 10), (4, 8)\}$

$$S = \{(2, 6), (4, 8)\}$$

$$R \cup S = R$$

$$R \cap S = S$$

$$R - S = \{(1, 2), (1, 6), (1, 8), (1, 10), (2, 2), (2, 8), (2, 10)\}$$

$$S - R = \emptyset$$

Example:

Let R_1 be the less than relation on the set of real numbers and let R_2 be the greater than relation on the set of real numbers, that is, $R_1 = \{(x, y) \mid x < y\}$ and $R_2 = \{(x, y) \mid x > y\}$. What are $R_1 \cup R_2$, $R_1 \cap R_2$, $R_1 - R_2$, $R_2 - R_1$, and $R_1 \oplus R_2$?

Solution:

We note that $(x, y) \in R_1 \cup R_2$ if and only if $(x, y) \in R_1$ or $(x, y) \in R_2$.

Hence, $(x, y) \in R_1 \cup R_2$ if and only if $x < y$ or $x > y$.

Because the condition $x < y$ or $x > y$ is the same as the condition $x \neq y$, it follows that

$R_1 \cup R_2 = \{(x, y) / x \neq y\}$. In other words, the union of the less than relation and the greater than relation is the not equals relation.

Next, note that it is impossible for a pair (x, y) to belong to both R_1 and R_2 because it is impossible that $x < y$ and $x > y$. It follows that $R_1 \cap R_2 = \emptyset$.

We also see that

$$R_1 - R_2 = R_1,$$

$$R_2 - R_1 = R_2,$$

$$\text{and } R_1 \oplus R_2 = R_1 \cup R_2 - R_1 \cap R_2 = \{(x, y) / x \neq y\}.$$

- **Composition of Relation**

Let R and S be two relations from sets A to B and B to C respectively, then we can define relation $S \circ R$ from A to C such that $(a, c) \in S \circ R \Leftrightarrow \exists b \in B$ such that $(a, b) \in R$ and $(b, c) \in S$.

This relation $S \circ R$ is called the composition of R and S .

Example:

Consider the sets $A = \{a, b\}$, $B = \{0, 1, 2\}$, and $C = \{x, y\}$.

The relation R between sets A and B is given by $R = \{(a, 0), (a, 2), (b, 1)\}$.

The relation S between sets B and C is defined as

$$S = \{(0, x), (0, y), (1, y), (2, y)\}.$$

The composition of relations $S \circ R$ is written as

$$S \circ R = \{(a, x), (a, y), (b, y)\}.$$

Example: What is the composite of the relations R and S , where R is the relation from $\{1, 2, 3\}$ to $\{1, 2, 3, 4\}$ with $R = \{(1, 1), (1, 4), (2, 3), (3, 1), (3, 4)\}$ and S is the relation from $\{1, 2, 3, 4\}$ to $\{0, 1, 2\}$ with $S = \{(1, 0), (2, 0), (3, 1), (3, 2), (4, 1)\}$?

Solution: $S \circ R$ is constructed using all ordered pairs in R and ordered pairs in S , where the second element of the ordered pair in R agrees with the first element of the ordered pair in S .

For example, the ordered pairs $(2, 3)$ in R and $(3, 1)$ in S produce the ordered pair $(2, 1)$ in $S \circ R$.

Computing all the ordered pairs in the composite, we find

$S \circ R = \{(1, 0), (1, 1), (2, 1), (2, 2), (3, 0), (3, 1)\}$.

Figure below illustrates how this composition is found.

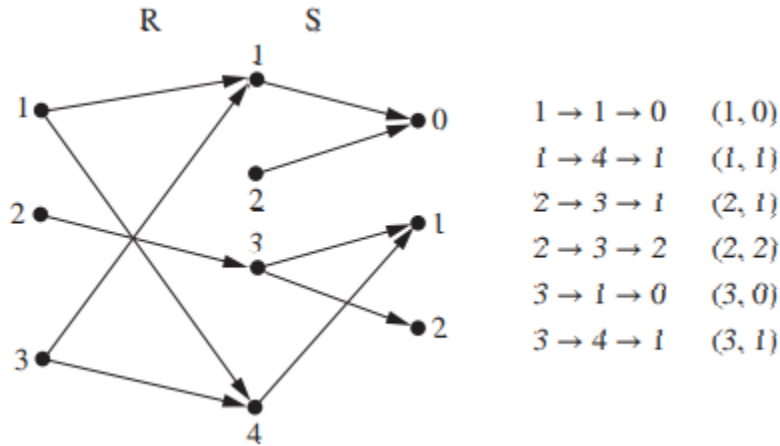


Figure: $S \circ R$

Reversal rule:

(i) $R \circ S \neq S \circ R$

(ii) $(S \circ R)^{-1} = R^{-1} \circ S^{-1}$

Example

Given the set $A = \{0, 1, 2\}$ with the relations $R = \{(0, 1), (0, 2), (1, 1), (2, 0)\}$ and $S = \{(0, 0), (0, 2), (1, 0), (1, 1)\}$. Find the compositions $S \circ R$ and $R \circ S$.

Solution:

$$S \circ R = \{(0, 0), (0, 1), (1, 0), (1, 1), (2, 0), (2, 2)\}$$

$$R \circ S = \{(0, 0), (0, 1), (0, 2), (1, 1), (1, 2)\}$$

Example:

Let $R = \{(1, 2), (1, 6), (2, 4), (3, 4), (3, 6), (3, 8)\}$

$S = \{(2, u), (4, s), (4, t), (6, t), (8, u)\}$

Find $S \circ R$.

Solution.

$$S \circ R = \{(1, u), (1, t), (2, s), (2, t), (3, s), (3, t), (3, u)\}$$

Example

Given the set $A = \{a, b, c\}$ with the relations $R = \{(a, a), (a, c), (b, a), (c, b)\}$ and $S = \{(a, b), (b, c), (c, c)\}$. Find the composition $S^{-1} \circ R^{-1}$.

Solution:

First we write the inverse relations R^{-1} and S^{-1} :

$$R^{-1} = \{(a, a), (c, a), (a, b), (b, c)\} = \{(a, a), (a, b), (b, c), (c, a)\};$$

$$S^{-1} = \{(b, a), (c, b), (c, c)\}.$$

$$S^{-1} \circ R^{-1} = \{(a, a), (b, b), (b, c)\}.$$

- **Powers of Binary Relations**

If a relation R is defined on a set A , it can always be composed with itself. So, we may have

$$R \circ R = R^2,$$

$$R \circ R \circ R = R^3 = R^2 \circ R$$

$$R \circ R \circ R \circ R = R^4 = R^3 \circ R$$

In general

$$R \circ R \circ \dots \circ R = R^{n+1} = R^n \circ R$$

Example:

Let $R = \{(1, 1), (2, 1), (3, 2), (4, 3)\}$. Find the powers R^n , $n = 2, 3, 4$.

Solution: Because $R^2 = R \circ R$, we find that $R^2 = \{(1, 1), (2, 1), (3, 1), (4, 2)\}$.

Furthermore, because $R^3 = R^2 \circ R \rightarrow R^3 = \{(1, 1), (2, 1), (3, 1), (4, 1)\}$.

Additional computation shows that $R^4 = R^3 \circ R$ is the same as R^3 ,

so $R^4 = \{(1, 1), (2, 1), (3, 1), (4, 1)\}$.

Example

The relation S on the set of real numbers R is defined by $S = \{(x, y) / y = x^2 + 1\}$.

Find the composition of relations S^2

Solution:

The composition S^2 is given by the property:

$$S^2 = S \circ S = \{(x, z) / \exists y \in S: xSy \wedge ySz\},$$

where

$$xSy = \{(x, y) / y = x^2 + 1\}, \quad ySz = \{(y, z) / z = y^2 + 1\}.$$

We eliminate the variable y in the second relation by substituting the expression $y = x^2 + 1$ from the first relation:

$$z = y^2 + 1 = (x^2 + 1)^2 + 1 = x^4 + 2x^2 + 2.$$

So we have:

$$S^2 = \{(x, z) / z = x^4 + 2x^2 + 2\}.$$

- **Partial Order Relation**

Definition: A relation $\leq$ on a set A is called a **partial order** if $\leq$ is reflexive, antisymmetric, and transitive. In this case we call A a **poset**.

Example:

Show that the set $\mathbb{Z}$ of integers together with the relation of inequality $\leq$ is a poset.

Solution:

1. $\leq$ is reflexive: For all $x \in \mathbb{Z}$ we have $x \leq x$ since $x = x$.
2. $\leq$ is antisymmetric: By the law of real numbers, for a given pair of numbers x and y only one of the following is true: $x < y$, $x = y$, or $x > y$.
So if $x \leq y$ and $y \leq x$ then we must have $x = y$.
3. $\leq$ is transitive: By the transitivity property of $<$ in Real numbers (R)
if $x < y$ and $y < z$ then $x < z$.
Thus, if $x \leq y$ and $y \leq z$ then the definition of $\leq$ and the above property imply that $x \leq z$.

Example:

Show that the relation $a|b$ in Natural numbers (N) is a partial order relation.

Solution:

- a) Reflexivity: Since $a = 1 \cdot a$, we have $a|a$.
- b) Antisymmetry: Suppose that $a|b$ and $b|a$. Then there exist positive integers k_1 and k_2 such that $b = k_1 a$ and $a = k_2 b$. Hence, $a = k_1 k_2 a$ which implies that $k_1 k_2 = 1$.
Since $k_1 k_2 \in N$, we must have $k_1 = k_2 = 1$; that is, $a = b$.
- c) Transitivity: Suppose that $a|b$ and $b|c$. Then there exist positive integers k_1 and k_2 such that $b = k_1 a$ and $c = k_2 b$. Thus, $c = k_1 k_2 a$ which means that $a|c$.

Example:

Let A be a collection of subsets. Let R be the relation defined by $A R B \Leftrightarrow A \subseteq B$.
Show that A is a poset.

Solution:

- I. $\subseteq$ is reflexive: For any set $X \in A$, $X \subseteq X$.
- II. $\subseteq$ is antisymmetric: By the definition of $\subseteq$
if $X \subseteq Y$ and $Y \subseteq X$ then $X = Y$, where $X, Y \in A$.
- III. $\subseteq$ is transitive: We have seen in the Sets that if $X \subseteq Y$ and $Y \subseteq Z$ then $X \subseteq Z$.

- **Total Order relations:**

Definition: if A is a **poset** then we say that a and b are **comparable** if either $a \leq b$ or $b \leq a$. If every pair of elements of A are comparable then we call $\leq$ a **total order** relation.

Example:

Consider the “divides” relation defined on the set $A = \{5, 15, 30\}$.
Prove that this relation is a total order on A .

Solution:

The fact that the “divides” relation is a partial order is easy to verify. Since $5|15$, $5|30$, and $15|30$, any pair of elements in A are comparable. Thus, the “divides” relation is a total order on A .

Example:

Show that the “divides” relation on N is not a total order.

Solution:

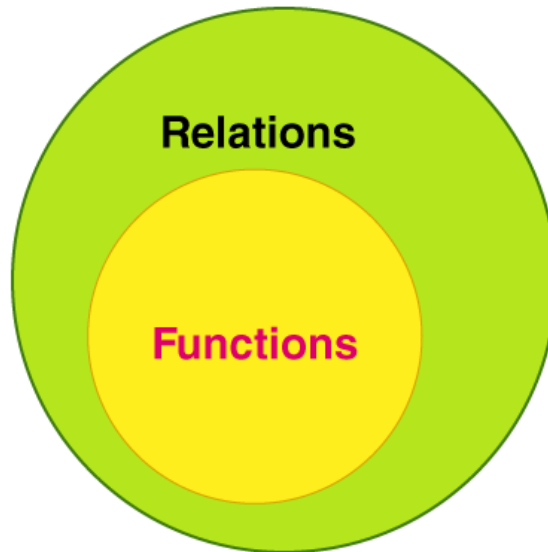
A counterexample of two noncomparable numbers are 2 and 3, since 2 does not divide 3 and 3 does not divide 2.

Important Results on Relation

- If R and S are two equivalence relations on a set A , then $R \cap S$ is also on ‘equivalence relation on A .
- The **union** $R \cup S$ of two equivalence relations on a set is **not necessarily** an equivalence relation on the set.
- If R is an equivalence relation on a set A , then R^{-1} is also an equivalence relation on A .
- If a set A has n elements, then number of reflexive relations from A to A is 2^{n^2-2} .
- Let A and B be two non-empty finite sets consisting of m and n elements, respectively. Then, $A \times B$ consists of mn ordered pairs. So, total number of relations from A to B is 2^{mn} .

3.1.2 Functions

All functions are relations, but not all relations are functions.



How to convert a Relation into a function?

A function is a special kind of relation (a set of ordered pairs) which follows a rule *i.e.* every x -value should be associated with only one y -value is called a Function.

Definition of Function:

- A relation is a correspondence between two variables.
- A relation can be expressed through a set of ordered pairs, a graph, a table, or an equation.
- A set containing ordered pairs (x, y) defines y as a function of x if and only if no two ordered pairs in the set have the same x -coordinate. In other words, every input map to exactly one output.
- A function is a rule that assigns each input exactly one output.

We call:

- the output the image of the input.
- The set of all inputs for a function is called the domain.
- The set of all allowable outputs is called the codomain.

We would write:

- $f: X \rightarrow Y$ to describe a function with name f , domain X and codomain Y .
- Also, we write $y = f(x)$ and say “ y is a function of x .”

For the function defined by $y = f(x)$,

- x is the independent variable (also known as the input)
- y is the dependent variable (also known as the output)
- f is the function name

In general:

If f is a function from A to B ($f: A \rightarrow B$) (i.e. f maps A to B), we say that A is the **domain** of f and B is the **codomain** of f .

If $f(a) = b$, we say that b is the *image* of a and a is a *preimage* of b .

The **range**, or *image*, of f is the set of all images of elements of A and is a subset of Codomain B consisting of values of f , (i.e. $\text{im}(f) = \{f(a) : a \in A\}$).

For example: $f: \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $f(x) = 3x$.

The **domain** and **codomain** are both the **set of integers**. However, the **range** is only the **set of integer multiples of 3**.

The rule which specifies a function can come in many different forms. Most often it is a formula, as in the $f(x) = 3x$ example of the previous paragraph.

Sometimes you need a few formulas, as in $g: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$g(x) = \begin{cases} 2x & x < 0 \\ x^2 & x \geq 0 \end{cases}$$

domain of g = all real numbers ($\mathbb{R}$).

Functions which are defined by different formulas on different intervals are sometimes called ***piecewise defined functions***.

Example: Find the domain and range of each of the following functions:

1. $f(x) = 8$

This is a constant function and so any value of x that we plug into the function will yield a value of 8. This means that the range is a single value or,

Range: 8 and the domain is all real numbers ($\mathbb{R}$) (i.e. $-\infty < x < \infty$ or $(-\infty, \infty)$).

2. $f(x) = 5x - 3$

We know that this is a line and that it's not a horizontal line (because the slope is 5 and not zero...). This means that this function can take on any value and so the range is all real numbers. i.e. $(-\infty, \infty)$.

3. $\boxed{g(t) = \sqrt{4 - 2t}}$

This is a square root and we know that square roots are always positive or zero. We know then that the range will be, $\mathbb{R}^+ = [0, \infty)$.

For the domain: We need to make sure that we don't take square roots of any negative numbers, so we need to require that,

$$4 - 2t \geq 0 \Rightarrow 4 \geq 2t \Rightarrow 2 \geq t \Rightarrow t \leq 2 \Rightarrow \text{Domain: } t \leq 2 \text{ or } (-\infty, 2]$$

4. $f(x) = |x - 6| - 3$

This function contains an absolute value and we know that absolute value will be either positive or zero. In this case the absolute value will be zero if $x = 6$ and so the absolute value portion of this function will always be greater than or equal to zero. We are subtracting 3 from the absolute value portion and so we then know that the range will be, Range: $[-3, \infty)$.

We can plug any value into an absolute value and so the domain is once again all real numbers or, Domain = $\mathbb{R} = -\infty < x < \infty$ or $(-\infty, \infty)$.

5. $h(x) = -3x^2 + 12x + 5$

In this case the range requires a little bit of work. From an Algebra class we know that the graph of this will be a parabola that opens down (because the coefficient of the x^2 is negative) and so the vertex will be the highest point on the graph. If we know the vertex we can then get the range. The vertex is then,

$$x = -\frac{12}{2(-3)} = 2, \quad y = h(2) = -2(2)^2 + 12(2) + 5 = 7 \Rightarrow (2, 7)$$

So, as discussed, we know that this will be the highest point on the graph or the largest value of the function and the parabola will take all values less than this, so the range is then, Range: $(-\infty, 7]$

Example: Find the domain and range of the following rational functions:

i. $f(x) = \frac{2}{\sqrt{4x-12}}$ Domain = $(3, +\infty)$ and range = $\mathbb{R}^+ = (0, +\infty)$

ii. $g(x) = \frac{x^2}{x^2+6}$ Domain = $\mathbb{R} = (-\infty, +\infty)$ and range = $[0, 1]$

iii. $h(x) = \frac{4}{x+1}$ Domain = $\mathbb{R} / \{-1\}$ and range = $\mathbb{R} / \{0\}$

iv. $h(x) = \frac{x+3}{x^2+3x+10} = \frac{x+3}{(x+5)(x-2)}$ Domain = $\mathbb{R} / \{-5, 2\}$ and range = $\mathbb{R} / \{0\}$

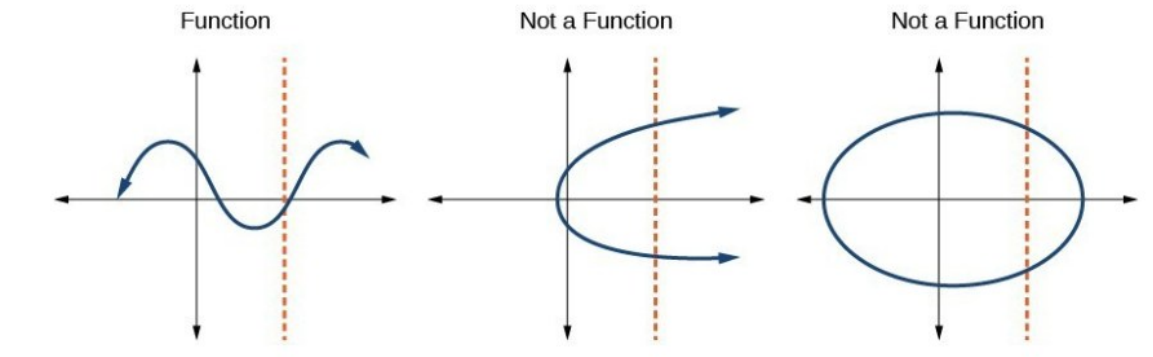
v. $h(x) = \frac{\sqrt{10x-5}}{x^2-16}$ Domain = $[1/2, +\infty) / \{4\}$ and range = $\mathbb{R} = (-\infty, +\infty)$

More examples in the lecture

Vertical Line Test

The vertical line test is a graphical method of determining whether a curve in the plane represents the graph of a function by visually examining the number of intersections of the curve with vertical lines.

The motivation for the vertical line test is as follows: A relation $f: A \rightarrow B$ is a function precisely when each element $x \in A$ is matched to at most one value $f(x) \in B$ and, as a result, **any vertical line** in the plane can intersect the graph of a function at most once. Therefore, the vertical line test concludes that a curve in the plane represents the graph of a function if and only if no vertical line intersects it more than once.



3.1.3 Injections, Surjections and Bijections

We now turn to investigating special properties functions might or might not possess.

3.1.3.1 Definition: Injective or one-to-one Function:

Let $f: A \rightarrow B$ be a function. We say that f is **injective** or **one-to-one** if and only if for all $x, y \in A$; if $f(x) = f(y)$ then $x = y$.

Using the concept of **contrapositive**, a function f is injective if and only if for all $x, y \in A$, if $x \neq y$ then $f(x) \neq f(y)$.

Taking the **negation** of this last conditional implication we see that f is not injective if and only if there exist two distinct elements a and b of A such that $f(a) = f(b)$.

Horizontal Line Test:

Once you have determined that a graph defines a function, an easy way to determine if it is a one-to-one function is to use the **horizontal line test**. Draw horizontal lines through the graph. If any horizontal line crosses the graph more than once, then the graph does not define a one-to-one function

Example:

- (a) Show that the function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 2x+3$ is an injective function.
- (b) Show that the function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^2 - 1$ is **not** an injective function.

Solution:

- (a) Suppose that $f(x_1) = f(x_2)$, then $2x_1 + 3 = 2x_2 + 3$ and this implies that $x_1 = x_2$.

Hence, f is an injective.

- (b) f is not injective since $f(-2) = f(2)$ but $-2 \neq 2$. Hence, f is **not** an injective

Example:

Show that if $f: \mathbb{R} \rightarrow \mathbb{R}$ is increasing then f is an injective (one-to-one).

Solution:

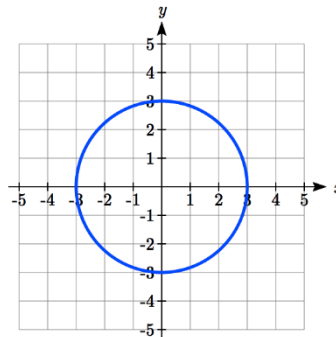
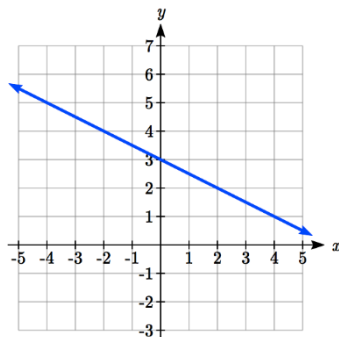
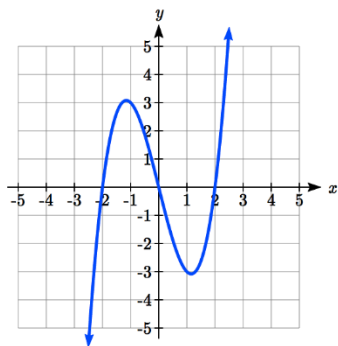
Suppose that $x_1 \neq x_2$. Then without loss of generality we can assume that $x_1 < x_2$.

Since f is increasing, $f(x_1) < f(x_2)$. That is, $f(x_1) \neq f(x_2)$.

Hence, f is an injective (one-to-one).

Example:

Which of these graphs defines a function $y=f(x)$? Which of these graphs defines a one-to-one function?



Graph 1 is not a one-to-one function. For example, the output value 3 has two corresponding input values, -1 and 2.3

Graph 2 is a one-to-one function; each input corresponds to exactly one output, and every output corresponds to exactly one input.

Graph 3 is not even a function so there is no reason to even check to see if it is a one-to-one function.

3.1.3.2 Definition: Surjective or onto Function

Now, for any function $f: A \rightarrow B$ we have $\text{Range}(f) \subseteq B$.

If equality holds then we say that f is **surjective** or **onto**.

It follows from this definition that a function f is **surjective** if and only if for each $y \in B$ there is an $x \in A$ such that $f(x) = y$.

i.e if every $y \in B$ has a preimage. In this case, the range of f is equal to the codomain

By taking the **negation** of this we see that f is not onto if there is a $y \in B$ such that $f(x) \neq y$ for all $x \in A$.

Example:

(a) Show that the function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 2x+3$ is surjective.

(b) Show that the function $f: \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $f(n) = 2n+3$ is not surjective.

Solution:

(a) Let $y \in \mathbb{R}$. Is there an $x \in \mathbb{R}$ such that $f(x) = y$? That is, $2x+3 = y$.

But solving for x we find $x = (y+3)/2 \in \mathbb{R}$ and $f(x) = y$. Thus, f is **onto**.

(b) Take $m = 4$. If f is onto then there should be an $n \in \mathbb{Z}$ such that $f(n) = 4$.

That is, $2n+3 = 4$: Solving for n we find $n = 1/2$ which is not an integer.

Hence, f is **not onto**.

3.1.3.3 Definition: Bijective Function

we say that a function f is **bijective** if and only if f is **both injective and surjective**.

Example:

Show that the function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 2x+3$ is a bijective.

Solution:

We can see from above two examples that $f(x) = 2x+3$ is an injective and surjective so, it is a bijective.

More examples in the lecture

3.1.4 Combining Real-valued Functions

A lot of the time the functions we encounter in mathematics are real-valued, *i.e.* functions whose **domain and co-domain are subsets of the real numbers $\mathbb{R}$** .

Combining functions means performing basic arithmetic (addition, subtraction, multiplication, and division) with functions.

Let's start with basic arithmetic of functions. Given two functions $f(x)$ and $g(x)$ we have the following notation and operations.

$$(f + g)(x) = f(x) + g(x)$$

$$(f - g)(x) = f(x) - g(x)$$

$$(f \cdot g)(x) = f(x)g(x)$$

$$(\lambda f)(x) = \lambda (f(x)), \lambda \in \mathbb{R}$$

$$(1/f)(x) = 1/f(x) \text{ or in general } (\lambda/f)(x) = \lambda/(f(x)), \lambda \in \mathbb{R}$$

$$(f/g)(x) = f(x)/g(x)$$

Sometimes we will drop the (x) part and just write the following,

$$f + g = f(x) + g(x), f - g = f(x) - g(x), f \cdot g = f(x)g(x), \lambda f = \lambda (f(x)), \lambda \in \mathbb{R}$$

$$(1/f) = 1/f(x) \text{ or in general } (\lambda/f) = \lambda/(f(x)), \lambda \in \mathbb{R} \text{ and } f/g = f(x)/g(x)$$

There is one new way of combining functions that we'll need to look at as well.

3.1.5 Function Composition

Now we need to discuss the new method of combining functions. The new method of combining functions is called **function composition**.

Composing functions is the process of applying functions in sequence, one after the other. You will see it occurring in all areas of mathematics. It is a fundamental mathematical concept and we can study it even before admitting any notion of number or measurement etc. It will play an important role in our study of groups in the next chapter.

Definition: Let $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ be two functions (note that the co-domain of f is also the domain of g). Then we can form a new function

$$g \circ f: X \rightarrow Z,$$

defined by the formula $(g \circ f)(x) = g(f(x))$.

We can refer to $g \circ f$ as g composed with f , or more informally, g *after* f

Example: Consider the real-valued functions f and g defined by the formulas

$$f(x) = 2 + 3x - x^2 \text{ and } g(x) = 2x - 1$$

Then we can find formulas defining the composition as follow,

$$(f \circ g)(x) = f(g(x)) = f(2x - 1) = 2 + 3(2x - 1) - (2x - 1)^2 = 2 + 6x - 3 - 4x^2 + 4x - 1 = -4x^2 + 10x - 2$$

$$(g \circ f)(x) = g(f(x)) = g(2 + 3x - x^2) = 2(2 + 3x - x^2) - 1 = 4 + 6x - 2x^2 - 1 = -2x^2 + 6x + 3$$

Note that these formulas are different. In general, even when both compositions are defined, composition is not commutative, *i.e.* $f \circ g \neq g \circ f$, although in special cases they may be equal.

More examples in the lecture

3.1.5.1 Definition: (Identity function)

The identity function ***id*** on the domain X is the function

$$id : X \rightarrow X,$$

defined by the formula

$$\forall x \in X \text{ } id(x) = x.$$

We shall speak of the identity function in different contexts without necessarily specifying the domain. The domain can always be determined from the specific context. The identity function has the property that when composed with another function, it leaves the other function unchanged, i.e.

$$f \circ id = id \circ f = f.$$

Sometimes the effect of a function can be reversed, as in the following definition.

3.1.5.2 Definition: (Inverse function)

Let $f : X \rightarrow Y$ be a function that is **bijective**, i.e. is injective and surjective. Then there exists a function

$$f^{-1} : Y \rightarrow X,$$

called the inverse of f , which has the property that

$$f^{-1} \circ f = f \circ f^{-1} = id.$$

Or in other words,

$$\forall x \in X \text{ } f^{-1}(f(x)) = x,$$

and

$$\forall y \in Y \text{ } f(f^{-1}(y)) = y.$$

Now, be careful with the notation for inverses. The “-1” is NOT an exponent even though it sure does look like one! When dealing with inverse functions we have got to remember that

$$f^{-1}(x) \neq \frac{1}{f(x)}$$

This is one of the more common mistakes that students make first studying inverse functions.

3.1.5.3 Finding the Inverse of a Function:

Given the function $f(x)$ we want to find the inverse function $f^{-1}(x)$

1. First, replace $f(x)$ with y .
2. Replace every x with y and replace every y with an x .
3. Solve the equation from step 2 for y (i.e. make y as a subject).
4. Replace y with $f^{-1}(x)$.

You can verify your answer by checking that

$$(f \circ f^{-1})(x) = x = id(x) \text{ and } (f^{-1} \circ f)(x) = x = id(x)$$

Example 1: Given $f(x) = 3x - 2$, find $f^{-1}(x)$.

Solution:

1. First, replace $f(x)$ with $y \rightarrow y = 3x - 2$
2. Replace every x with y and replace every y with an $x \rightarrow x = 3y - 2$
3. Solve the equation from step 2 for y (i.e. make y as a subject) $\rightarrow 3y = x + 2 \rightarrow y = (x + 2)/3$

4. Replace y with $f^{-1}(x) \Rightarrow f^{-1}(x) = (x+2)/3$

Example 2: Given $g(x) = \sqrt{x-3}, x \geq 0$, find $g^{-1}(x)$.

Solution:

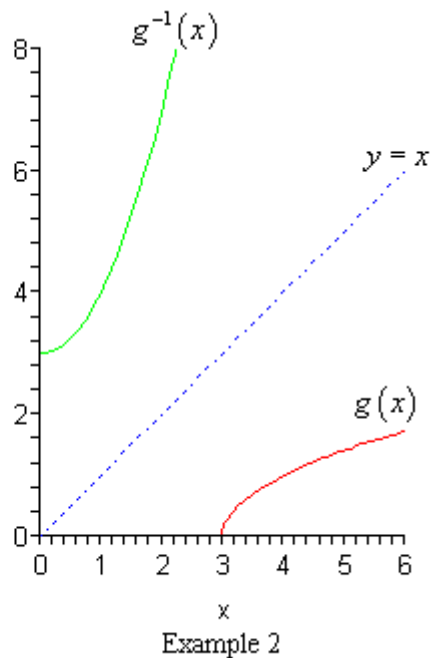
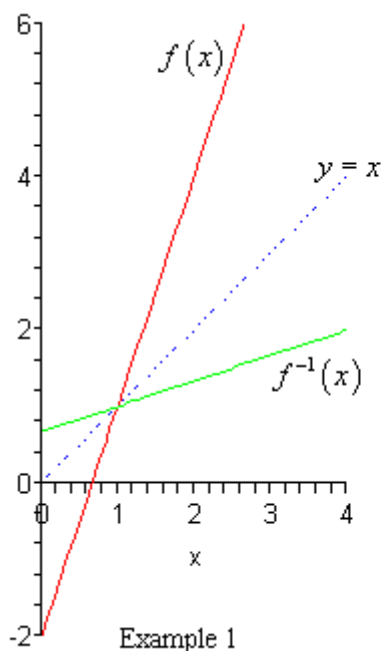
1. First, replace $g(x)$ with $y \Rightarrow y = \sqrt{x-3}$

2. Replace every x with y and replace every y with an $x \Rightarrow x = \sqrt{y-3}$

3. Solve the equation from step 2 for y (i.e. make y as a subject) $\Rightarrow x^2 = y-3 \Rightarrow y = x^2 + 3$

4. Replace y with $g^{-1}(x) \Rightarrow g^{-1}(x) = x^2 + 3$

Below is the graph of the function and inverse for example 1 & 2. In both examples we can see that the graph of the inverse is a reflection of the actual function about the line $y=x=id(x)$. This is always the case with the graphs of the function and its inverse.



Theorem: Suppose that $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are two **bijective** functions. Then the composition $g \circ f$ is also **bijective** and

$$(g \circ f)^{-1} = f^{-1} \circ g^{-1}.$$

3.2 Rational Functions and their Partial Fraction Expansion:

Rational functions are those functions that are quotients of polynomial functions. So a rational function $f(x)$, is a real-valued function of the form

$$f(x) = \frac{P(x)}{Q(x)}$$

where both $P(x)$ and $Q(x)$ are polynomials, and the degree of $P(x)$ is smaller than the degree of $Q(x)$. Recall that the degree of a polynomial is the largest exponent in the polynomial. So, let's do a quick review of partial fractions.

Partial fractions can only be done if the degree of the numerator is strictly **less than** the degree of the denominator. That is important to remember. So, once we've determined that partial fractions can be done, we factor the denominator as completely as possible. Then for each factor in the denominator we can use the following table to determine the term(s) we pick up in the partial fraction decomposition.

Factor in denominator	Term in partial fraction decomposition
Linear Factor $(ax+b)$	$\frac{A}{ax+b}$ Ex) $\frac{-x+1}{x^2+3x+2} = \frac{A}{x+1} + \frac{B}{x+2}$
A repeated linear factors $(ax+b)^k$	$\frac{A_1}{ax+b} + \frac{A_2}{(ax+b)^2} + \dots + \frac{A_k}{(ax+b)^k}, k=1,2,3,\dots$ Ex) $\frac{x^2-6x+3}{(x-2)^3} = \frac{A}{x-2} + \frac{B}{(x-2)^2} + \frac{C}{(x-2)^3}$ Ex) $\frac{5x^2+21x+4}{(x+1)^2(x-3)} = \frac{A}{(x+1)^2} + \frac{B}{x+1} + \frac{C}{x-3}$
Quadratic factors ax^2+bx+c	$\frac{Ax+B}{ax^2+bx+c}$ Ex) $\frac{2x+1}{x^2+3x+4}$
Distinct Linear and Quadratic factors	Ex) $\frac{x^2+4x-23}{(x^2+4)(x+3)} = \frac{Ax+B}{x^2+4} + \frac{C}{x+3}$

A Repeated
Quadratic Factor
 $(ax^2 + bx + c)^k$

$$\frac{A_1x + B_1}{ax^2 + bx + c} + \frac{A_2x + B_2}{(ax^2 + bx + c)^2} + \dots + \frac{A_kx + B_k}{(ax^2 + bx + c)^k}, k = 1, 2, 3, \dots$$

Ex) $\frac{-3x^3 - x}{(x^2 + 1)^2} = \frac{Ax + B}{x^2 + 1} + \frac{Cx + D}{(x^2 + 1)^2}$

There are several methods for determining the coefficients for each term and we will go over each of those in the following examples.

Example 1 Express $\frac{3x + 11}{x^2 - x - 6}$ as the sum of its partial fractions.

Solution

The first step is to factor the denominator as much as possible and get the form of the partial fraction decomposition. Doing this gives,

$$\frac{3x + 11}{(x - 3)(x + 2)} = \frac{A}{(x - 3)} + \frac{B}{(x + 2)}$$

The next step is to actually add the right side back up.

$$\frac{3x + 11}{(x - 3)(x + 2)} = \frac{A(x + 2) + B(x - 3)}{(x - 3)(x + 2)}$$

Now, we need to choose A and B so that the numerators of these two are equal for every x . To do this we will need to set the numerators equal.

$$3x + 11 = A(x + 2) + B(x - 3)$$

Note that in most problems we will go straight from the general form of the decomposition to this step and not bother with actually adding the terms back up. The only point to adding the terms is to get the numerator and we can get that without writing down the results of the addition.

At this point, we have one of two ways to proceed. One way will always work, but is often more work. The other, while it will not always work, is often quicker when it does work. In this case, both will work and so we will use the quicker way for this example. We will look at the other method in a later example.

What we are going to do here is to notice that the numerators must be equal for *any* x that we would choose to use. The numerators must be equal for $x=-2$ and $x=3$. So, let us plug these in and see what we get.

$$\begin{aligned} \text{For } x=-2 &\rightarrow 5=A(0)+B(-5) \rightarrow B=-1 \\ \text{For } x=3 &\rightarrow 20=A(5)+B(0) \rightarrow A=4 \end{aligned}$$

So, by carefully picking the x 's we got the unknown constants to quickly drop out. Note that these are the values we claimed they would be above.

$$\frac{3x+11}{x^2-x-6} = \frac{4}{x-3} - \frac{1}{x+2}$$

Example 2 Express $\frac{2x^2+3}{(x^2+x-2)}$ as partial fractions.

Solution

Since the degree of numerator is equal the degree of denominator then we must divide, so that,

$$\frac{2x^2+3}{x^2+x-2} = 2 + \frac{-2x+7}{x^2+x-2} = 2 + \frac{A}{(x+2)} + \frac{B}{(x-1)}$$

$$\frac{-2x+7}{x^2+x-2} = \frac{A}{(x+2)} + \frac{B}{(x-1)}$$

Using the same method above we get $A=11/3$ and $B=5/3$

$$\frac{2x^2+3}{(x^2+x-2)} \equiv 2 - \frac{11}{3(x+2)} + \frac{5}{3(x-1)}$$

Example 3 Express $\frac{3x-11}{(x^2+1)(x+3)}$ as partial fractions

Solution

$$\frac{3x-11}{(x^2+1)(x+3)} = \frac{Ax+B}{x^2+1} + \frac{C}{x+3}$$

$$3x-11 = (Ax+B)(x+3) + C(x^2+1)$$

$$\text{if } x=3 \Rightarrow C = -2$$

$$\text{if } x=0 \Rightarrow B = -3$$

$$\text{let } x=1 \Rightarrow A = 2$$

$$\text{So } \frac{3x-11}{(x^2+1)(x+3)} = \frac{2x-3}{x^2+1} - \frac{2}{x+3}$$

Exercise 5: Relations & Functions

- I. List the ordered pairs in the relation R from $A = \{0, 1, 2, 3, 4\}$ to $B = \{0, 1, 2, 3\}$, where $(a, b) \in R$ if and only if
- a) $a = b$. b) $a + b = 4$. c) $a > b$. d) $a \mid b$.
- II. For each of these relations on the set $\{1, 2, 3, 4\}$, decide whether it is reflexive, whether it is symmetric, whether it is antisymmetric, and whether it is transitive.
- a) $\{(2, 2), (2, 3), (2, 4), (3, 2), (3, 3), (3, 4)\}$
b) $\{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$
c) $\{(2, 4), (4, 2)\}$
d) $\{(1, 2), (2, 3), (3, 4)\}$
e) $\{(1, 1), (2, 2), (3, 3), (4, 4)\}$
f) $\{(1, 3), (1, 4), (2, 3), (2, 4), (3, 1), (3, 4)\}$
- III. Let R be the relation $\{(1, 2), (1, 3), (2, 3), (2, 4), (3, 1)\}$, and let S be the relation $\{(2, 1), (3, 1), (3, 2), (4, 2)\}$. Find $S \circ R$, S^{-1} , R^{-1} , $R^{-1} \circ S^{-1}$ and $(S \circ R)^{-1}$
- Write the domain and range of the following functions:
- IV. i) $f(x) = -|2x - 8|$, ii) $g(x) = \sqrt{x - 4} + 3$ and iii) $h(x) = \frac{5 + x}{x - 2}$
- V. For each of the functions below determine which of the properties hold, injective, surjective, bijective. Briefly explain your reasoning.
- i) The function $f : R \rightarrow R$ defined by $f(x) = e^x$.
ii) The function $f : R \rightarrow R^+$ defined by $f(x) = e^x$.
iii) The function $f : R \rightarrow R$ defined by $f(x) = x(x + 1)(x - 1)$.
iv) The function $f : Z \rightarrow Z$ defined by $f(x) = x - 5$
v) The function $f : [3, +\infty) \rightarrow [5, +\infty)$ defined by $f(x) = (x - 3)^2 + 5$
- 1) If $f(x) = x + 1$, $g(x) = 5x^4$ and $h(x) = -6$
Find the following:
- VI. i) $g(f(x))$, ii) $f(h(x))$ and iii) $(hog)(x)$
- 2) If $f(x) = 4 \log_3(x)$, $g(x) = 3^x$ and $h(x) = 9 - x$
Find the following:
- i) $(hofog)(x)$ and ii) $(fogoh)(x)$

Find the inverse of the following functions:

VII. $i) f(x) = 5x^3 - 6,$ $ii) g(x) = \sqrt{4x + 7},$

$iii) h(x) = \log_2(x) - 10$ $iv) k(x) = \frac{x+2}{x-4}$

VIII. Express the following rational functions in partial fractions:

1. $\frac{-x+1}{x^2+3x+2}$

2. $\frac{x^2-6x+3}{(x-2)^3}$

3. $\frac{5x^2+21x+4}{(x+1)^2(x-3)}$

4. $\frac{-3x^3-x}{(x^2+1)^2}$